

Chapter 43 — Exstrophy-Epispadias Complex

- **One spectrum of genitourinary malformation:** from glanular epispadias (mildest) through classic bladder exstrophy to cloacal exstrophy (an overwhelming multisystem defect).
- **Five entities** (Fig. 43.1): cloacal exstrophy; male classic bladder exstrophy; female classic bladder exstrophy; male epispadias; female epispadias.

HISTORICAL ASPECTS

- **Assyrian-Babylonian cuneiform tablets** (first–second millennia BCE) recorded birth anomalies as omens; genital anomalies were frequent, renal and bladder anomalies rare and uninterpretable — **a definitive description of bladder or cloacal exstrophy was not corroborated.**
- **First epispadias:** attributed to Byzantine emperor Heraclius (CE 610–641). **First cloacal exstrophy:** Schenck, 1595.

Incidence and Inheritance

- **Bladder exstrophy incidence:** 2.15 per 100,000 live births (United States); about 1 in 46,000 in an international prospective study. German live prevalence: whole complex about 1 in 11,000; exstrophy about 1 in 30,675.
- **Male-to-female ratio 2.3:1** across series; some series report 5–6:1.
- **Recurrence in a family about 1 in 100. Offspring of an affected parent: 1 in 70** live births — about 500-fold the general population. No affected offspring of affected parents were reported until the 1980s.
- **Twin data:** concordance is inconsistent even in monozygotic twins, so the cause is **multifactorial, not simple Mendelian**. Multiplex families and consanguinity occur but are rare.
- **Other anomalies in relatives** are mostly **midline defects and oral clefts**. Cleft lip and palate are also prevalent in probands.
- **Parental factors:** older parents on average; many probands are first pregnancies, which may reflect later first pregnancy and more use of assisted reproduction.
- **Phenotype correlates** (a multicentre study of mild, intermediate/classic and severe/cloacal forms): males over-represented in every group; **maternal first-trimester smoking → cloacal exstrophy; preconception folic acid → mildest phenotype (epispadias).**
- **Hormonal exposure:** high-dose early first-trimester progesterone (about tenfold rise) and in vitro fertilisation (IVF; about 7.5-fold rise) both increase risk, which implicates a hormonal role. In a large female cohort, parental age, reproductive history, alcohol, drugs, radiation and infection showed no association, but periconception smoking was more common in cloacal exstrophy.

Genetics

- **CASPR3 (chromosome 9):** breakpoint disruption in the 5' region — the first suggestion of a genetic basis.
- **22q11.2 duplication:** the **most common genetic variant associated with bladder exstrophy–epispadias complex (BEEC)**. No single coding gene within the locus is duplicated in isolation.
- **Risk loci on seven chromosomes.**
- **p63 (p53 family):** the ΔNp63 isoform drives stem-cell proliferation and inhibits apoptosis, and is implicated in development of the abdominal and ventral bladder wall. It is under-expressed and dysregulated in exstrophic urothelium, but **no genomic p63 mutation has been found.**

- **Expression profiling:** candidate genes cluster in cellular assembly, musculoskeletal development and connective tissue; desmosomal/cytoskeletal genes are largely under-expressed. The **most downregulated genes are desmin and desmuslin**, which interact with desmoplakin (overexpressed).
- **ISL1 (5q11.1):** a LIM/homeobox transcription factor, confirmed by genome-wide association as a **susceptibility gene for classic bladder exstrophy (CBE)**. It regulates genital tubercle factors (Bmp4, Fgf10, Wnt5a). Variants may impair its transcription and its interaction with oestrogen receptor α .

Concept — *The epidemiology points to a multifactorial, partly heritable and hormone-sensitive field defect: familial clustering and a single dominant locus both fail to explain it, while environmental modifiers (smoking, folate, IVF) shift where on the spectrum a fetus lands.*

EMBRYOLOGY

- **Core defect:** the **cloacal membrane fails to be reinforced by mesodermal ingrowth**. This bilaminar membrane at the caudal germinal disc occupies the infraumbilical wall. Mesenchyme growing between its ectoderm and endoderm normally forms the lower abdominal muscles and pelvic bones.
- **Normal sequence:** mesenchymal ingrowth → urorectal septum grows caudally and divides the cloaca (bladder anterior, rectum posterior) → the septum meets the membrane remnant, which perforates to form the urogenital and anal openings → paired genital tubercles migrate medially and fuse cephalad to the membrane (Fig. 43.2, 4–16 mm stages).
- **Timing and extent of rupture decide the variant. Classic exstrophy is more than 50% of the complex.** Cloacal exstrophy is the earlier defect of the same developmental field.

THEORIES OF EXSTROPHY EMBRYOGENESIS

Theory	Proposed defect	Consequence
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Timing of rupture → bladder exstrophy, cloacal exstrophy or epispadias
Lateral body-wall fold failure	One or both folds fail to meet in the midline	Failure in abdomen and pelvis → cloacal; pelvis alone → classic
Caudal genital hillocks	Tubercles fuse below rather than above the membrane	Dorsal urethral defect
Abnormal body-stalk insertion	Caudal insertion stops midline mesenchyme interposing; cloaca not translocated deep	Superficial, unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Pelvic-ring structures cannot meet → bladder herniates

- **Pattern:** every theory converges on **absent infraumbilical midline mesoderm**. They differ only in whether the primary event is in the membrane, the body wall, the genital tubercle or the pelvis.

Concept — *Missing midline mesoderm below the umbilicus is the single lesion; the site and timing of membrane rupture explain why one field defect produces three phenotypes.*

CLASSIC BLADDER EXSTROPHY

KEY POINTS: THE EXSTROPHY-EPISPADIAS COMPLEX

- In CBE, most anomalies involve the **abdominal wall, bladder, genitalia, pelvic bones, rectum and anus**.
- The **male genital defect is severe** and remains the most troublesome part of reconstruction.

- In females the **vagina is short (hardly more than 6 cm) but of normal calibre**, and the cervix lies in the anterior vaginal wall near the introitus. The orifice is often stenotic and displaced anteriorly, the clitoris is bifid, and the labia, mons and clitoris diverge.

Anatomic Considerations

Bony Pelvis and Spinal Defects

- **Three-dimensional (3D) computed tomography (CT)** showed that the pelvis is not simply widened at the symphysis.
- **Rotational anomalies:** external rotation of the posterior pelvis/ilic wings (mean 12° each side) → external rotation of the anterior segment (mean 18° each side) → coronal rotation of the sacroiliac joint (about 10° larger than normal; pelvis about 15° more inferiorly rotated) → acetabular retroversion → convergence of the iliac wings → femoral retroversion.
- **Dimensional anomalies:** increased pubic diastasis; **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%. The **posterior segment length is normal**. The sacrum has larger volume and surface area.
- **Clinical consequences:** the rotation contributes to the **short, pendular penis**; lateral displacement of the innominate bones causes wide hips, a waddling gait and outward rotation of the limbs. These cause little disability and improve with growth (foot progression angle 20–30° beyond normal early on).
- **Bone histology and endochondral ossification are normal** in fetal exstrophy pelvises, so the 30% shortfall is unexplained. Restoring a closed pelvic ring may normalise growth and load distribution.
- **Cloacal exstrophy:** greater deformity, plus right–left asymmetry, sacroiliac malformation and occasional hip dislocation.
- **Spine:** clinically insignificant variants (spina bifida occulta, lumbarisation or sacralisation) are fairly common; scoliosis and dysraphism (myelomeningocele, lipomeningocele, scimitar sacrum, hemivertebrae) are uncommon; **neurological dysfunction is exceptional**.

Pelvic Floor Defects

- **Puborectal sling supports twice the normal cavity area.** The levator ani is **displaced posteriorly** (most of it behind the rectum), rotated outward, and flattened from its normal cone into a flat, less dome-shaped sheet. **Muscle length and thickness are normal** (Fig. 43.5: paucity of anterior pelvic floor musculature before closure).
- **Levator curvature does not track the size of the diastasis.**
- **Postclosure imaging:** continent patients have the shortest intersymphyseal distance, the most normal levator divergence and the deepest bladder neck. In dry adults the **puborectalis angle is under 65°**.
- **Closure reshapes the pelvis** from a box into an inwardly rotated hammock → shifts levators into the anterior compartment → smooths the floor contour. This supports **aggressive posterior placement of the vesicourethral unit**, osteotomy and pelvic fixation.

Abdominal Wall Anomalies

- **Triangular fascial defect** containing the exstrophied bladder and posterior urethra. It is **bounded inferiorly by the intrasymphyseal band** (the divergent urogenital diaphragm), which tethers the posterior vesicourethral unit to the pubic ramus. The anterior rectus sheath fans behind the urethra and bladder neck to insert into this band.
- **No striated sphincter** is found in the band. Bladder muscle extends laterally and interdigitates with rectus fascia to form a fibrous urogenital diaphragm. **Radical incision of these fibres down to the levator hiatus is essential**; they are often found intact at reclosure of failed repairs.

- **Umbilicus** sits at the apex, well below the iliac crest line, with a **shortened umbilicus-to-anus distance**. Umbilical hernia, if present, is small and repaired at closure. Omphalocele is rare in CBE (common in cloacal exstrophy).
- **Indirect inguinal hernia** is common, especially in males, and often bilateral. Causes: patent processus vaginalis, large rings, lack of canal obliquity. **Repair at closure** (excise the sac and repair the transversalis fascia and muscle) and **explore the contralateral side**. Osteotomy at closure lowers both hernia incidence and recurrence.

Anorectal Defects

- **Short, broad perineum**; the **anus is displaced anteriorly**, directly behind the urogenital diaphragm, at the posterior limit of the fascial defect.
- **Colorectal anomalies about 2%** — low, but about 72-fold the general population. Commonest: **imperforate anus → rectal stenosis → congenital rectal prolapse**.
- The divergent levators and distorted external sphincter give **imperfect early anal continence**.
- **Rectal prolapse**: before closure, prolapse is transient, reducible, **resolves with closure and pubic apposition**, and is an **indication to proceed with closure**. Prolapse **after** closure → **suspect posterior urethral or outlet obstruction → urgent cystoscopy**.

Male Genital Defect

- **Anterior corporal length about 50% shorter** than controls; **posterior corporal length normal**; anterior corporal diameter about 30% greater. The corpora separate in **parallel**, so the intercorporal angle is unchanged (Fig. 43.7).
- The penis therefore looks short because of **both the pubic diastasis and a true congenital deficiency of anterior corporal tissue**.
- **Surgical anatomy**: triangular, separated corpora; a long convex ventral surface and a short wedge-shaped dorsal surface; the lie of the neurovascular bundle on each corpus sets its length.
- **Prostate**: in infants, larger transverse and anteroposterior dimensions and volume. In adults, **normal volume and weight but never circumferential** — the **urethra lies anterior to the prostate** (Fig. 43.8: intersymphyseal fibrous band runs along the whole prostate). Periprosthetic vessels and sensory bundles are externally rotated but follow normal landmarks.
- **Prostate-specific antigen (PSA)** is measurable but **below the age-specific upper limit**.
- **Vasa, ejaculatory ducts and seminal vesicles are normal**.
- **Cavernous nerves are displaced laterally** and almost always preserved (potency is retained). **Retrograde ejaculation** may follow closure or bladder neck reconstruction.
- **Testes**: fertility is not thought to be impaired by testicular dysfunction (harvested sperm for IVF had normal parameters). Testes often appear undescended but are **mostly retractile, with enough cord length to reach the scrotum without orchidopexy**.

Female Genital Defects

- **Simpler reconstruction than in the male** (Fig. 43.9). The short vagina, anterior cervix, stenotic anterior orifice, bifid clitoris and divergent labia are as in the Key Points. **Tubes and ovaries are normal**.
- **At primary closure**: join the clitoral halves, and join the labia minora to form a fourchette. In adulthood, **vaginal dilation or episiotomy** may be needed for intercourse.
- **Levator hiatus**: normal anteroposterior length but **almost twice normal width**. This predisposes to **uterine prolapse** — usually after childbirth but also in nulliparous women (mean age of onset in one series 16 years), sometimes needing suspension.

- **Osteotomy is not protective; adult diastasis width is the only predictor of prolapse.**
- **Clitoral body lies mostly anterior to its pelvic attachment** in females with CBE, unlike normal females and males with CBE (Fig. 43.10).

Urinary Defects

- **Newborn mucosa is usually normal** but may carry hamartomatous polyps. **Irrigate with saline and cover with a protective membrane** to prevent cystic or metaplastic change.
- **Size, distensibility and neuromuscular function** of the plate, and the size of the fascial defect, decide whether repair is attempted.
- **The newborn exstrophic bladder is immature, not neuropathic:**
 - **Muscarinic receptors:** normal density and affinity.
 - **Collagen-to-smooth-muscle ratio:** raised (type III collagen increased threefold, type I unchanged). It **falls after successful infection-free closure** and approaches normal by adulthood, and stiffness falls with it.
 - **Myelinated nerves:** reduced through loss of small fibres, with large fibres preserved.
 - **Immunohistochemical neural markers** (vasoactive intestinal polypeptide, neuropeptide Y, substance P, calcitonin gene-related peptide, protein gene product 9.5, NADPH diaphorase): no dysinnervation.
 - → **a maturational delay with potential for normal development after successful closure.**
- **Small, fibrotic, inelastic, polyp-covered plate → functional repair may be impossible** (Fig. 43.11). Polyps are fibrotic or oedematous; about half carry **squamous metaplasia**; von Brunn nests, cystitis cystica and **cystitis glandularis** occur, the last more often in secondary closures → **adenocarcinoma risk → urine cytology and cystoscopy into adulthood**. Polyps after a failed closure impede growth.
- A bladder that is invaginated or bulges through a small defect signals good potential capacity. **True size is judged only under anaesthesia**, because pelvic depth cannot otherwise be assessed.
- **Urodynamics:** most continent patients have normal cystometrograms, and most bladders are compliant and stable before bladder neck reconstruction. **After bladder neck reconstruction compliance and stability fall**, and only a minority keep normal detrusor function. Series disagree, probably because **bladder neck techniques are not standardised**.
- **Ultrastructure** (Fig. 43.12): **caveolae** (cell-cell signalling) are normal after successful closure but **deficient in bladders that later need augmentation**; failed closures show collagen deposition and degenerating muscle.
- **Smooth-muscle cell biology:** actin and myosin are normal. Exstrophy cells migrate more, driven by **transforming growth factor- β 1 (TGF- β 1)** and not by calcium; proliferation is equal. Findings on intracellular calcium and contractile force conflict between studies. **Delayed-closure cells show less proliferation (Ki67) and more autophagy and apoptosis.**
- **Urothelium:** squamous metaplasia, inflammation and loss of transitional differentiation. **Uroplakin-II (the barrier plaque protein) is reduced, most in delayed and repeat closures**, and metaplasia is commoner after delayed closure → **argues for early closure**. Even after successful infant closure only a minority show a fully differentiated phenotype (uroplakin 3a, cytokeratins 13 and 20, claudin 4). Tight-junction barrier formation in culture is variable, and squamous lines usually fail — a possible transcriptional/genetic basis.
- **Upper tracts usually normal**; renal anomalies are uncommon (commonest: **duplex system**; also hypoplastic or absent, pelvic, ureteropelvic junction obstruction, multicystic dysplastic), but more frequent than in the general population.

- **Ureteric course:** an enlarged, deep pouch of Douglas forces the ureter laterally and down, so it enters from below and lateral with **little or no obliquity** → **vesicoureteric reflux (VUR) in 100% of closed bladders** → **reimplantation at bladder neck reconstruction**.
- **Excess outlet resistance** at closure → recurrent infection despite prophylaxis → reimplantation and/or outlet revision may be needed before bladder neck reconstruction.
- **Staged reconstruction does not harm renal function.**

Concept — *Every anatomical abnormality traces to an open, externally rotated pelvic ring with an anteriorly displaced floor: short penis, wide hips, anterior anus, flat levators, oblique-less ureters and prolapse all follow, which is why closing the ring deep in the pelvis is the recurring surgical theme. The bladder itself is immature, not defective — its fate is set by what happens after birth.*

Exstrophy Complex Variants

- All variants share part of the **skeletal, urinary and genital triad**, consistent with a common embryological origin.

EXSTROPHY COMPLEX VARIANTS

Variant	Defining feature	Management / outcome
Skin-covered (split symphysis) — commonest	Full musculoskeletal defect, urinary tract essentially intact; bladder may show through thin skin; sometimes an isolated ectopic bowel segment (colon or ileum) on the lower wall, and only epispadias in males	Formal exstrophy closure at birth
Duplicate exstrophy (anteroposterior)	Everted mucosal patch on the wall plus a closed pelvic bladder that carries the ureters (patch is dry)	Excise ectopic mucosa and close the wall
Side-by-side duplication	Left and right bladders with a muscular septum, each with its own ureter and intact sphincter ; pubic and rectus diastasis common	—
Superior vesical fissure	Classic musculoskeletal defect; membrane ruptures only at its top → vesicostomy-like fistula above the umbilicus	Sphincter intact → toilet-trains normally, no bladder neck procedure
Pseudoexstrophy	Musculoskeletal defect only; elongated low umbilicus; recti attach to separated pubes (mesodermal migration interrupted superiorly only)	—
Epispadias with bladder prolapse	Classed with penopubic/complete epispadias but structurally close to exstrophy; prolapse may appear only on Valsalva	Prolapse repair ± osteotomy; diagnosis often delayed , especially in girls (at toilet training)

- **Outcome in a large series:** most variants underwent primary closure (often with osteotomy); many needed continence procedures, some with augmentation; volitional voiding was achieved in a minority. External genitalia vary (normal penis to complete epispadias).
- **Trap:** superior vesical fissure and skin-covered cloacal exstrophy do better than classic cases, and their inclusion in series created the **false belief that all variants do better**. **All variants except superior vesical fissure → formal closure at birth**, followed up like the classic form.

Prenatal Diagnosis

- **Often missed** or mislabelled as omphalocele or gastroschisis. **Only about 25% of BEEC is diagnosed prenatally**, even with 3D ultrasound and fetal magnetic resonance imaging (MRI).
- **Additional quantitative signs:** umbilical cord insertion-to-genital tubercle distance **below the 5th centile** for gestation; **pubic diastasis over 10 mm**.

- **Fetal MRI is reserved** for when 3D ultrasound cannot separate classic from cloacal exstrophy, or other severe anomalies (including spinal) are suspected.
- **Purpose:** counselling, and **delivery in a specialised exstrophy centre** for immediate reconstruction and multidisciplinary and psychological support.

KEY POINTS: PRENATAL DIAGNOSIS (ULTRASOUND CRITERIA FOR CBE)

- Absence of bladder filling
- Low-set umbilicus
- Widening pubic rami
- Diminutive genitalia (difficult to sex the fetus)
- Lower abdominal mass enlarging as pregnancy progresses (Fig. 43.14)

SURGICAL RECONSTRUCTION OF BLADDER EXSTROPHY

- **Evolution:** 1950s staged approach (iliac osteotomies 4–6 days before closure; separate epispadias repair; intrasymphyseal band wrapped round the urethra) → 1970s **three stages:** closure → bladder neck reconstruction with antireflux → epispadias repair.
- **Modern staged repair of exstrophy (MSRE):** newborn bladder, abdominal wall and posterior urethral closure well onto the penile shaft ($\pm$ bilateral innominate and vertical iliac osteotomy) → **epispadias repair at 6–12 months** → **bladder neck reconstruction with antireflux at 4–7 years**, once capacity is adequate and the child can join a voiding programme.
- **Alternatives:** combined closure with penile repair in the newborn (complete primary repair); staged repair without osteotomy using radical soft-tissue mobilisation (Kelly); total newborn repair (Erlangen); ureterosigmoidostomy with closure (Mainz).

Evaluation and Management at Birth

- **Mucosa is smooth but easily denuded.** Sequence: tie the cord with **2-0 silk close to the wall** (a clamp traumatises mucosa) → cover with non-adherent **plastic wrap** → at each nappy change remove it, irrigate with sterile saline and replace it (Fig. 43.15). Keep it covered during transfer.
- **Counselling** starts prenatally, from a surgeon experienced in exstrophy. The team includes paediatric orthopaedics, paediatric anaesthesia, social work, specialist nurses and a psychiatrist or psychologist. Parent support association available.
- **Cardiopulmonary assessment** in the first hours; **renal ultrasound suffices** before closure. Modern transport keeps any neonate within hours of a full centre.

KEY POINTS: BLADDER EXSTROPHY — EVALUATION OF THE NEWBORN

- Size and quality (including polyp burden) of the bladder template — **3 × 3 cm is suitable for newborn closure**
- Pubic diastasis (radiographic) and **pelvic malleability under anaesthesia** → need for osteotomy
- Urethral plate length and width; penile length
- Associated anomalies

Selection of Patients for Immediate Closure

- **An experienced exstrophy surgeon decides at birth.** Detrusor size and functional capacity drive success, but **apparent size is not potential capacity**: a small plate that bulges on crying or indents easily under anaesthesia may be adequate, and extra bladder may be found behind the fascia.
- Once relieved of surface trauma, a small bladder enlarges under minimal outlet resistance. **Robust infant + good template + experienced team → strongly consider newborn closure.**
- **Uroplakin-II is higher after newborn than delayed closure** → early closure may potentiate bladder growth.

The Inadequate Bladder Template Unsuitable for Newborn Closure

- **Inadequate** = **under 3 cm diameter**, polyp-covered, or inelastic/fibrotic. Examination under anaesthesia is often needed because oedema, excoriation and polyps develop after birth.
- **Conditions precluding primary closure**: penoscrotal duplication; hypoplastic bladder; significant bilateral hydronephrosis; ectopic bowel within the plate (**relative** contraindication).
- **Delayed primary closure (DPC) for small templates**: all such patients had osteotomy and closure succeeded. Continence procedures were as frequent and as successful as in controls, but the first continence procedure came later, continence came later, and a **catheterisable channel was needed far more often**. Capacities were somewhat smaller, but **annual growth rates were similar**.
- **Longitudinal data**: delayed closure gives lower capacities, with the steepest losses at 4–6 months and beyond 9 months. Short-term complications (transfusion, infection, dehiscence) rise slightly but are manageable.
- **Timing**: close **as early as possible**; if a delay is needed for growth, close at **6–9 months**. Waiting 6 months or more is safer than closing a small plate badly (dehiscence → future incontinence). Delay, when indicated, does not jeopardise growth.
- **No adequate growth by 6–12 months** → excise the bladder and use a **non-refluxing colon conduit or ureterosigmoidostomy**.

Pelvic Osteotomy

- **Mean pubic diastasis 4.8 cm**, an open pelvic ring. **Larger template → wider diastasis → greater need for osteotomy**. Hip function is normal; external foot rotation lessens with growth even without osteotomy.
- **Aims**: close the ring, relieve tension on the abdominal closure, improve later reconstruction, and **reduce primary and recurrent inguinal hernia**. First described in 1958 (bilateral posterior iliac osteotomy 1–3 weeks before closure), it lowered dehiscence rates.
- **When**: at any stage if the diastasis blocks the urological goals.
 - **Newborn**: not routine unless the **diastasis is over 4 cm or the pelvis is poorly malleable** (judged jointly by a senior paediatric urologist and a paediatric orthopaedic surgeon, under anaesthesia).
 - **Closure after 72 hours of age → osteotomy**.
 - **Under 72 hours with malleable pubes** that meet on medial rotation of the greater trochanters → closure without osteotomy. **If in doubt, osteotomise**.
 - Sacroiliac ligament laxity allows tension-free closure in the first 48–72 hours.
- **Newborn osteotomy is safe**: orthopaedic complications are rare. Newborns with osteotomy fail at a rate similar to carefully selected newborns without it.
- **Always required** after a failed closure with wide diastasis, in large templates and in cloacal exstrophy. **Extreme diastasis (over 6 cm)** may need **staged pelvic closure** after osteotomy.

- **Advantages at initial closure:** easy symphyseal approximation with less abdominal tension and no fascial flaps → vesicourethral unit placed **deep in the pelvic ring** (more outlet resistance) → pelvic floor muscles brought to the midline to support the bladder neck. Some patients then gain dry intervals or full continence.
- **Most referred dehiscences and prolapses had no osteotomy** at initial closure.
- **Operating time:** osteotomy plus closure takes 5–7 hours in an infant.

OSTEOTOMY TYPES COMPARED

Type	Position	Features
Combined bilateral anterior innominate + vertical iliac — most used	Supine (no turning)	Fixator pins placed under direct vision; greenstick closing wedge beside the sacrum gives two mobile fragments; better cosmesis; best malleability and tension-free pubic apposition; better approximation and less recurrence of diastasis than posterior
Bilateral posterior iliac	Prone (turn required)	Original technique; worse approximation and more recurrence
Transverse anterior innominate alone	Supine	Less pubic mobility than the combined approach
Superior pubic ramotomy; diagonal iliac-wing osteotomy	—	Less effective mobilisation

- **Success rates (institutional series):** in newborn closures, success did not differ by osteotomy type. In delayed closures, combined osteotomy was the most successful, anterior innominate intermediate and posterior iliac the least. Delayed closure with osteotomy succeeded more often than newborn closure in that series.
- **Radiological result:** repair reduces pelvic volume and anteverts the acetabula towards normal. Pelvic closure moves the levator ani anteriorly and strengthens the puborectalis sling.

Combined osteotomy technique: supine; prepare below the costal margins; gauze over the bladder → expose from iliac wings to pectineal tubercle and back to the sacroiliac joints → elevate periosteum and sciatic notch → **Gigli saw transverse innominate cut** exiting midway between the anterior superior and anterior inferior spines (slightly higher than a Salter osteotomy, to take pins) → osteotome **vertical closing-wedge cut just lateral to the sacroiliac joint**, leaving the posterior iliac cortex as a hinge → **two pins in the inferior segment and two in the iliac wing** (one each in newborns with little cancellous bone) → radiograph to confirm → urological repair → pubic suture → apply external fixator (Figs. 43.16–43.17).

- **Aftercare:** radiograph at 7–10 days; if the diastasis is not fully reduced, **crank the fixator bars** over several days. **Longitudinal (modified Buck) traction about 4 weeks; fixator 4–6 weeks** until callus forms (Fig. 43.18); pins removed at the bedside under light sedation.
- **Closure without osteotomy (under 48–72 hours): modified Bryant traction** (hips flexed 90°) for 4 weeks.
- **Pubic suture:** **No. 2 nylon horizontal mattress** between the fibrocartilage of the rami, tied anteriorly. Every approximation method is weaker than an intact symphysis, but this is the **strongest load-to-failure**.

Complications of Osteotomy and Immobilisation Techniques

- **Inadequate immobilisation** → failed closure, bladder prolapse, loss of suprapubic tubes and stents. Pain and movement control are equally important: **tunnelled epidural for 2–3 weeks** while callus forms.
- **Combined osteotomy complications:** **transient left femoral nerve palsy** (full recovery by 12 weeks; none on the right despite the same surgeon; managed with bed rest then a knee immobiliser), delayed iliac union, wound and pin-site infection, transient abductor weakness and peroneal palsy. **Pin-site inflammation is almost universal**, especially at the iliac crest, and is controlled with oral antibiotics.

- **Partial recurrence of diastasis** even after osteotomy, by two mechanisms: (1) **early pin loosening** before healing, mostly in infants (denser bone in older children gives more rigid fixation); (2) **undergrowth of the ischiopubic segment** (a third smaller in adults) with growth. Uncorrected diastasis widens with growth, so correction is still net positive. Classic exstrophy gains more correction than cloacal.
- **The main role of osteotomy is to relieve tension** on the bladder, posterior urethra and abdominal wall during healing. It is used less in newborns (ligament laxity) and is **essential for failed repairs, cloacal exstrophy, and newborns with a wide diastasis and an excellent template**. In combined closure with epispadias repair it lets the corpora be brought over the urethra.
- **Repeat osteotomy at reclosure is safe**: all reclosures succeeded, gait was usually normal and there were no nerve palsies.
- **Failures still occur after osteotomy**. On multivariate analysis failure is **reduced by Buck traction with external fixation for more than 4 weeks and increased when a general (not paediatric) orthopaedic surgeon performs the osteotomy**. Osteotomy type and epidural use do not matter.
- Some centres use spica casts (earlier discharge); spica advocates now apply them for 8 weeks.

TABLE 43.1 PELVIC IMMOBILIZATION AFTER THE CLOSURE OF CLASSIC BLADDER EXSTROPHY (FAILURE RATE, %)

Method	All closures	With osteotomy	Without osteotomy
Spica cast	47	30	66
External fixation + traction	7.5	8	0 (very few cases)

Modified from Baumgartner T, Jayman J, Friedlander D, et al., Society for Pediatric Urology Fall Congress, Montreal, 2017. All differences significant.

- **What it tests**: immobilisation method matters independently of osteotomy. **Spica without osteotomy fails in about two-thirds**; external fixation with traction fails in under 1 in 10.
- **Trap**: assuming osteotomy alone secures closure — **spica still fails in almost a third even after osteotomy**.
- **Recommendation**: **longitudinal traction plus external fixation**, especially with osteotomy.

IMMOBILISATION TECHNIQUES

- Mummy wrap
- Spica cast
- **Modified Bryant traction — only when osteotomy is not performed**
- **Modified Buck traction — with the external fixator**
- External fixator device

Concept — Osteotomy and immobilisation serve one goal: **a tension-free pelvic ring that holds while the closure heals**. Age (ligament laxity), diastasis width and fixation rigidity decide how much bony help is needed, and failures cluster where tension or movement was allowed.

Surgical Options in the Newborn With Classic Bladder Exstrophy

- **Anaesthesia: combined general and epidural**. Catheter advanced 6–10 cm to **T10–T12**. Load with 1 mL/kg 0.25% bupivacaine with 1:200,000 adrenaline in fractions, with electrocardiographic monitoring for intravascular injection. Top up every 2 hours with 0.6 mL/kg 0.25%, or 1 mL/kg 0.125%, with adrenaline.

PRIMARY RECONSTRUCTION STRATEGIES COMPARED

Repair	Stages and timing	Osteotomy	Distinguishing feature	Weakness
MSRE	Newborn closure → epispadias 6–12 mo → bladder neck 4–7 y	If >72 h, diastasis >4 cm or stiff pelvis	Converts exstrophy to complete epispadias; balanced outlet stimulates growth	Multiple operations
Complete primary repair of exstrophy (CPRE, Mitchell)	Newborn closure + penile disassembly + bladder neck tailoring	If >3 days or excessive diastasis	Aims for continence without formal bladder neck reconstruction	Hypospadias; soft- tissue/glans loss; early reimplantation
Kelly (radical soft-tissue mobilisation)	Dome closure + hernia repair at birth → 3–6 mo mobilisation, penoscrotal urethrostomy → hypospadias repair about 3 y	None (the purpose is to avoid it)	Pelvic floor muscles, pudendal vessels and nerves freed off ischium/ pubis and wrapped round the urethra	Fistula, stenosis
Schrott-Erlangen	“Total” repair at 8 weeks if adequate plate; if small, bladder-only closure first	None (only cloacal exstrophy and reoperations)	Everything in one sitting: — closure, reimplants, bladder neck plasty, groin exploration, epispadias repair, pubic closure; epidural 5 days	—
Mainz	Newborn closure → ureterosigmoidostomy at 2 years with penile/genital reconstruction	—	Remnant bladder becomes a seminal receptaculum; later Mainz sigma pouch lowers colonic pressure; oral alkalinisation	Colonic malignancy risk

Kelly Repair

- Developed to **avoid osteotomy** and improve on an earlier soft-tissue technique. The neourethra is made from the urethral plate dissected off the corpora (as in penile disassembly), but the **meatus is always placed at the penoscrotal junction first** and moved distally later.
- **Single-stage delayed closure with Kelly mobilisation** without osteotomy: all closed without dehiscence, but fistula or stenosis was common. **Kelly plus anterior component separation** can close large defects without osteotomy.

Complete Repair: Complete Primary Repair of Exstrophy

- **Aim:** fewer procedures, and continence without formal bladder neck reconstruction.
- **Penile disassembly** (Fig. 43.24): penis separated into **right corpus with hemiglans, left corpus with hemiglans, and urethral wedge** (plate with its spongiosum). Sequence: ventral start (as in Cantwell-Ransley) → medially above Buck fascia down to tunica albuginea → preserve spongiosum → carry posteriorly to the bladder neck → urethra between the corpora into a hypospadiac position → corpora sutured in the midline → hypospadias-style skin closure → proximal dissection into the pelvis for concomitant closure, plus **bladder neck tailoring**.
- **Now often modified** — the plate is **left attached to the glans** (resembling modified Cantwell-Ransley) because of soft-tissue loss after complete disassembly.
- **Females: more urinary retention when CPRE is done after 72 hours** (longer plate, narrower neck; the urethra is compressed at pubic approximation, risking ischaemia). Delayed CPRE → a gradual tapered transition with a wider bladder neck and a shorter plate.

- **Perineal membrane dissected off the pubis** to set the unit deep — a principle common to all good repairs.
- **Drawbacks: most boys are left hypospadiac; about half need ureteric reimplantation in the first year. Simultaneous bilateral ureteric reimplantation (BUR-CPRE)** is feasible and reduces postoperative VUR and later ureteric surgery, but is unproven against pyelonephritis.

Schrott-Erlangen Approach

- **The most evolved single-stage approach** (details in the table). Long-term (about 20-year) follow-up: most objectively continent, with mostly normal renal function.

Mainz Repair

- Began with failed closures and small plates, then applied to **all newborns from 1964** regardless of template. In girls: genital reconstruction plus **anterior uterine fixation**. Usually only one later cosmetic operation.
- **Principles common to every method:** (1) **radical mobilisation of the posterior vesicourethral unit**; (2) combined closure with epispadias repair only in very carefully selected patients; (3) **tension-free closure** of the abdominal wall and pelvis, with osteotomy if needed; (4) **strict selection criteria** for newborn closure.

Modern Staged Reconstruction of Bladder Exstrophy

- **Objective:** convert exstrophy into **complete epispadias** with the urethra on the proximal or mid shaft. **Balanced posterior outlet resistance** preserves the kidneys and stimulates bladder growth.
- **Epispadias repair about 6 months, after testosterone; bladder neck repair at 4–5 years** with adequate capacity and, above all, readiness for a voiding programme. Europe is trending towards staged repair.
- **Trap:** Erlangen, Kelly and CPRE were claimed to avoid bladder neck repair, but **almost all patients still need an outlet procedure**.

Bladder, Posterior Urethral, and Abdominal Wall Closure

Sequence (Fig. 43.20): outline a **2-cm-wide mucosal strip** from the distal trigone to below the verumontanum and on to the mid-shaft (male) or to the vaginal orifice (female) → incise above the umbilicus, along the bladder-skin junction, down to the plate → plane between rectus fascia and bladder; **umbilical vessels doubly ligated**; peritoneum stripped off the dome so the bladder sits deep → continue caudally to the urogenital diaphragm; a **double-pronged skin hook in the pubis pulled laterally** exposes the fibres → **divide the urogenital diaphragm fibres sharply with cautery down to the levator hiatus** → mobilise the plate up to the prostate if left in continuity → free the corpora from the suspensory ligaments and inferior rami (subperiosteal detachment of the urogenital diaphragm) → close mucosa and muscle of bladder, neck and urethra onto the mid-shaft → **taper the posterior unit over a 12-Fr sound** (meatus calibrated to 14 Fr) and buttress with a second layer → **suprapubic non-latex Malecot for 4 weeks; ureteric stents; urethra unstented** → pressure on the greater trochanters approximates the pubes → **No. 2 nylon horizontal mattress**, knot away from the neourethra, ± a second stitch at the caudal rectus insertion → **V-shaped skin flap forms the neumbilicus**, through which the tubes exit (Hanna's method).

- **Why incise the plate rarely:** only if it is so short that it limits penile length and causes dorsal angulation. Then lengthen it (Johnston or Duckett technique); for a very short penis, transect distal to the verumontanum and rotate paraexstrophy (non-hair-bearing) skin flaps distally, leaving the corpora un-united for the later Cantwell-Ransley repair.
- **Penile “lengthening” is apparent, not real** (anterior corpora are 50% short) — it comes from correcting chordee and angulation.

- **Tapering purpose:** narrow and lengthen the unit so it sits deep — **enough resistance to aid adaptation and prevent prolapse, not enough to harm the upper tracts.** This becomes the site of the later bladder neck repair.
- **Stents 10–14 days (up to 2–3 weeks):** closing a small bladder causes swelling that can obstruct the ureters and cause transient hypertension. The urethra is not stented because secretions accumulate and cause necrosis.
- **Broad-spectrum antibiotics** before and during surgery convert a contaminated field to a clean one.
- **The commonest error: inadequate dissection of the urogenital diaphragm.** Most failed closures, including repeatedly failed ones, have **intact fibres.** The unit is then pulled anteriorly when the pubes close, which increases prolapse risk and gives a poor position for later epispadias repair. **3D MRI-guided dissection** improves accuracy and safety (Figs. 43.21–43.22: fibres run along the medial and posteroinferior pubis and show glistening white on external rotation).

Combined Bladder Closure and Epispadias Repair

- First tried in the 1960s and **abandoned in the 1970s** for complications. Revived for failed closures and, later, for newborns. It combines closure with a **modified Cantwell-Ransley repair**, suits delayed closures and failed closures, and **does not create hypospadias.**
- **Limit to boys older than 6 months** because of the risk of **penile and corporal tissue loss.** Select by phallic size and length, depth of the urethral groove, template size (delayed closure) and perivesical and urethral scarring (failed closure). **Not for the occasional exstrophy surgeon.**

Management After Primary Closure

- MSRE leaves **mid-shaft epispadias with incontinence.** About **3 days in intensive care;** tunnelled epidural lidocaine plus patient-controlled analgesia.
- **At 4 weeks, before suprapubic tube removal:** calibrate the outlet (catheter or sound) → full ultrasound → clamp the tube to estimate residual → urine culture. **Continuous antibiotic prophylaxis, since all have VUR.**
- **Ultrasound surveillance:** 3 months after discharge, then every 6–12 months for 2–3 years. **Yearly cystoscopy and cystography under anaesthesia** for reflux and growth — capacity in an awake, crying infant is unreliable.
- **Capacity under 30 mL by 1–2 years → warn parents** that a continence procedure may not be feasible. **Best predictors: template size at birth plus a successful, infection-free primary closure.** A useful continence interval after closure alone is unusual.
- **Outlet too tight** (retention, reflux, dilatation, infected urine): urethral dilation or intermittent catheterisation → transurethral incision of a posterior stricture → early antireflux surgery (6–12 months) → surgical revision (skin-flap advancement or patch) only if these fail.
- **Recurrent urinary tract infection (UTI) plus distended bladder → cystoscopy** looking for **intrapubic stitch erosion** into the posterior urethra → remove it transurethrally or through a small suprapubic incision.
- **Upper tracts stay acceptable** with prophylaxis and **residuals kept under 50 mL.** Glomerular filtration rate (GFR) after bladder neck reconstruction is no worse than before closure.
- **Diversion is seldom needed** if an adequate outlet was built and follow-up is meticulous.

KEY POINTS: MANAGEMENT AFTER PRIMARY CLOSURE — KEYS FOR SUCCESS

- Adequate bladder drainage
- Adequate ureteric drainage
- Adequate immobilisation and movement control

- Adequate pain control

Selected Technical Aspects of Other Methods of Closure

Kelly Repair

Stage 1: close the bladder to the posterior urethra with **no pubic apposition**. **Stage 2 (radical soft-tissue mobilisation):** incise round the old closure with parallel cuts on the plate to halfway between the ureteric orifices and the verumontanum → excise lateral mucosa → expose the dorsal penile nerves and corporal attachments → move the plate beneath the corpora (penoscrotal hypospadias) → enter the extraperitoneal space behind the rectus and retract the vas → cut a **“flake” of superior pubis** carrying the corpora and urogenital diaphragm remnants → incise the levator through to its pubic insertion → free the pudendal vessels and nerves in the ischiorectal fossa → bring the urethral and bladder neck muscles together without tension (Fig. 43.23).

Mitchell Repair

- Often done after a few months of age. The urethra is freed from the corpora and pelvic diaphragm at the first stage, which Mitchell argues places the bladder neck deeper. Originally thought sufficient for continence — **most still need bladder neck repair**.
- The penis is disassembled into three parts on the premise that each corpus has its own blood supply, but the **plate depends on the spongiosum → ischaemia risk**. It is now modified to keep the plate attached to the glans.
- Intersymphyseal band incised medial to the inferior pubis down to the levator hiatus (as in MSRE) → bladder neck tailored → urethra taken towards the tip. **If it falls short (most patients), the meatus is left hypospadiac at the base.**
- **Females:** bladder neck, urethra and vagina mobilised **as one unit**.
- **Osteotomy (as MSRE, unlike Kelly) if over 3 days old or the diastasis is excessive.**

Penile and Urethral Closure in Exstrophy

Epispadias Repair

- **Order reversed historically:** bladder neck reconstruction used to precede urethroplasty. The discovery that **epispadias repair increases bladder capacity** (via outlet resistance) moved it earlier (Fig. 43.19: marked capacity gain after urethroplasty).
- **Now performed at 6–10 months** in all. Completion **before 12 months gives better eventual capacity**.
- **Testosterone before every urethroplasty** (small penis, scarce skin): **testosterone enanthate 2 mg/kg intramuscularly, 5 weeks and 2 weeks before**.
- **Current techniques:** Cantwell-Ransley (1989), modified Cantwell-Ransley (1995), Mitchell–Bägli penile disassembly (1996).
- **Four key concerns:** (1) correct dorsal chordee; (2) reconstruct the urethra; (3) reconstruct the glans; (4) close penile skin.
- **Formal elongation and chordee release are always needed at urethroplasty:** excise suspensory ligament remnants and old scar, and free the corpora further from the inferior rami (surprisingly little is achieved at initial closure). Gains are apparent, not true.
- **Complete disassembly:** in over two-thirds the plate does not reach the tip → hypospadias needing another operation.

Chordee

- **Options:** dorsomedial corporal **incision and anastomosis**; **dermal graft** lengthening of the dorsum. **Grafting is reserved for adolescents and adults** with residual chordee and a need for length.

Urethral Reconstruction

- **Abandoned methods:** tubularisation of the dorsal groove in place; free genital or extragenital skin grafts; ventral transverse island flaps; double-faced island flaps.
- **Modern principle:** tubularise the plate and **move it beneath the corpora** → fewer fistulae, a downward penile deflection and an easily catheterisable channel.
- **Hypospadias repair after disassembly:** reported as easy by some, but a large European series found a high complication rate. Better results came with a **two-stage repair with a graft**.
- **Preferred repair if the groove is long enough: modified Cantwell-Ransley** (urethra already closed well up the shaft at primary closure).

Modified Cantwell-Ransley sequence (Fig. 43.25): transverse glans traction stitch → parallel incisions outlining an **18-mm strip** from the prostatic meatus to the tip → **deep vertical incision of the distal plate closed transversely** with 6-0 polyglycolic acid (**reverse meatal advancement and glanuloplasty, reverse MAGPI**) flattens the plate and brings the meatus to the tip → excise dorsal glans mucosa beside the strip; raise thick glans wings → mobilise lateral skin flaps; a **suprapubic Z-incision** exposes and divides the suspensory ligaments and scar → take ventral skin down to the scrotum, **preserving the plate's mesentery** → dissect the corpora ventrally on Buck fascia until emerging dorsally between spongiosum and corpus on each side; vessel loops → free the plate from the corpora up to the prostate and distally past the glans-corporal junction, **leaving the distal 1 cm attached to the glans** (not complete disassembly) → neurovascular bundles left intact in young children if rotation straightens the penis, otherwise dissected free on loops → **tubularise over an 8-Fr silicone stent** (6-0) → if needed, **diamond-shaped corporotomies at maximum curvature**, sutured to each other → close corpora over the neourethra (5-0 polydioxanone), burying it further at the corona (4-0) → close the glans wings in two layers (5-0 subcuticular, 6-0 epithelium) → ventral skin to the corona; flaps resurface the dorsum → **Z-plasty at the base** → stent 10–12 days.

- **Result:** the urethra is displaced **ventrally** → downward deflection, with some added length from dorsal rotation of the corpora.

Penile Skin Closure

- **Z-plasty at the penile base** prevents contraction and upward tethering. Ventral foreskin is split into **lateral preputial flaps** brought dorsally (asymmetry staggers the suture line, which helps) or **buttonholed and transposed** to the dorsum.

Postoperative Problems

- **Bladder spasm control is paramount** — spasms cause extravasation and fistula. Caudal epidural plus **oxybutynin from the end of surgery**. Discharged with the plastic dressing intact, opiates, antispasmodics and broad-spectrum antibiotics.

Female Exstrophy

- **Principles:** (1) radical dissection of bladder and urethra **keeping the vaginal connection**; (2) free the lateral vesicourethral–vaginal unit from the urogenital diaphragm; (3) tension-free closure of abdomen, bladder–urethra and genitalia; (4) judicious osteotomy and proper immobilisation.
- **Osteotomy if pubes over 4 cm apart or not malleable.** The **diastasis is narrower in females** even with a large template → closure is easier and osteotomy is needed less often than in males.

Sequence: iodine vaginal preparation (a breach is simply closed) → dissect along the medial clitoral/corporal halves deep into the pelvis with low-setting pinpoint cautery → vagina laterally and posteriorly; **urethrovaginal septum left intact** (blood supply) → down to the levator hiatus, freeing the urogenital fibres, so the unit sits deep and is not pulled forward at pubic closure → bladder: **single figure-of-eight layer of 3-0 polyglactin** (maximises volume); urethra: single layer 4-0-5-0 → suprapubic tube via the dome plus two feeding-tube stents for **4 weeks**, exiting the neoumbilicus; **nothing through the neourethra** (infection, prolapse, dehiscence) → No. 2 nylon pubic mattress → skin → **monsplasty**; clitoral halves joined (5-0/6-0) → **Y-V labioplasty** if needed to exteriorise the introitus → fixator and radiograph → **Buck traction 4 weeks if osteotomised, Bryant 4 weeks if not.**

Continence and Antireflux Procedure

- Many patients arrive **still incontinent after newborn repair alone or after repeated bladder neck injections**; most groups recommend some outlet procedure.
- **Yearly cystoscopy and gravity cystogram under anaesthesia** after closure. Expected growth after newborn closure is about **54 mL over 24 months. Median capacity about 100 mL** is typical in those who end up completely dry; they are usually 5–7 years old and developmentally ready.
- **Pre-bladder-neck programme** (nurse practitioner, nurses, child psychologist) starts **at least 6 months before**; completing it is a prerequisite for surgery.
- **Reflux is managed conservatively** with prophylaxis until bladder neck reconstruction; **earlier reimplantation** for pyelonephritis, recurrent UTI, worsening hydronephrosis or falling renal function.

Modified Young-Dees-Leadbetter (YDL) sequence (Fig. 43.26): open by a **transverse incision at the bladder neck with a vertical extension** (the later midline closure lengthens the short exstrophy bladder) → **Cohen cross-trigonal or cephalotrigonal reimplantation**; if the orifices sit low, cut the hiatus cephalad and reimplant high on the trigone → select a **posterior mucosal strip 15 × 30 mm** from mid-trigone to prostate or posterior urethra → denude the lateral muscle triangles (adrenaline-soaked sponges for haemostasis) → multiple small edge incisions let the reconstruction ride cephalad; **no transverse cut at the cephalad end** (unlike the original Young-Dees), which avoids denervation and ischaemia → close mucosa and muscle over the strip (4-0) as a **mucosa-lined tube within a muscular funnel** → **double-breast the denuded flaps** (3-0 polydioxanone) → an 8-Fr stent only as a guide, then removed → **suspend to the rectus fascia.**

- **Very radical dissection**, including off the back of the pubic bar, is needed for adequate narrowing and anterior suspension. After a **CPRE with a vesicocutaneous fistula**, tissues adhere to the intrasymphyseal bar; the bar may be divided for exposure (repair with 2-0 polydioxanone) and **abduction then restricted** while it heals.

Postoperative Care

- Ureteric stents through the bladder wall; **suprapubic tube 3 weeks**; at **4 weeks clamp for no more than 1 hour** at first → no void → **8-Fr Foley under anaesthesia for 5 days** → repeat trial; several cycles may be needed. Remove the suprapubic tube once emptying is satisfactory. **Frequent bladder and renal ultrasound** in the first months.

Concept — *Staged reconstruction is a sequence of controlled outlet resistances: enough after closure to grow the bladder, more after urethroplasty to enlarge it further, and definitive only once capacity and the child's cooperation can use it. Every stage done out of order or too tight either stunts growth or threatens the kidneys.*

MODERN INITIAL REPAIR OF BLADDER EXSTROPHY: OUTCOMES AND RESULTS

- **Functional reconstruction transformed outcomes;** early newborn closure with or without osteotomy is well established, with acceptable continence and preserved renal function in most.
- **Two most reliable predictors of eventual continence: template size at birth and a successful primary closure.** Referred patients whose primary closure elsewhere succeeded mostly became continent after bladder neck repair by a second surgeon → **success of the primary closure matters more than who performed it.**

Initial Closure

- **Successful primary closure** (with or without osteotomy) brings **quicker and higher continence.**
- **Each failed closure erodes potential:** after two closures, well over half still reach adequate capacity but **fewer than 1 in 5** gain voided continence; after three, adequate capacity drops to half. **Most multiply-closed patients had no osteotomy** at their first closure.
- **Closure before 1 year:** far fewer need eventual augmentation than when closed later.
- **CPRE complications** are comparable in frequency to MSRE but **more serious:** fascial dehiscence, **vesicocutaneous fistula**, early reimplantation for upper-tract change (prompting calls for reimplantation at closure), **major prolapse or dehiscence even with osteotomy**, and **soft-tissue loss** — partial or complete glans loss, loss of the urethrovaginal septum.
- **Erlangen:** mild complications (occasional fistula, minimal hydronephrosis, rare severe hydronephrosis needing surgery), no prolapse or dehiscence, with no osteotomy but a sophisticated **obturator-foramen coaptation** technique.
- **Volume and training: high-volume hospitals (more than 5 closures a year) have lower costs.** A successful newborn closure costs far less than a reclosure. **Reclosures by fellowship-trained paediatric urologists** achieve better success and continence than those by general urologists, non-fellowship paediatric urologists or paediatric surgeons.

Epispadias Repair

- **Incontinence is the biggest problem, but genital appearance is the greatest concern for males;** many adolescents rate an odd-looking penis as worse than incontinence.
- **Modified Cantwell-Ransley outcomes (CBE and epispadias):** most penises horizontal or downward on standing; early fistula about 1 in 6, falling by 3 months; proximal anastomotic strictures and dorsal skin separation occasional; **a negotiable, catheterisable channel in all;** a few need further straightening; sexually active adults report orgasm and ejaculation with a straight erection. From 1994 the **plate is detached from the corona except the distal 0.5–1 cm within the glans**, to bury the urethra deeper.
- **Ransley's reverse MAGPI concept (1989):** dorsomedial corporal incision and anastomosis plus distal meatotomy. Long-term fistula and stricture rates are **low (about 5%).**
- **Why coronal fistula is rare:** dissecting the strip inside the glans places the urethra ventrally, beneath raw glans tissue apposed over it (unlike the Young repair). **Fistulae occur at the penile base**, where the urethra rises between the corpora. With scarce tissue (failed closure or repair, recurrent fistula) → **tunica vaginalis flap.**
- **Paraexstrophy skin flaps cause strictures** → reserved for when standard lengthening fails; most surgeons preserve the plate at closure.
- **Mitchell-Bägli disassembly:** fistula and stricture rates similar, but **a large share are made hypospadiac** (reports range roughly from over a third to over four-fifths). **Ischaemic loss of glans, plate and corpora** is reported; the largest series links it mainly to closure **without osteotomy** → pudendal vessel compression

from tension, or direct injury → **stop and perform an immediate osteotomy** to restore flow. Hypospadias repair afterwards can be difficult → modified disassembly keeps the plate attached distally.

- **No repair adds true corporal length**, not even plate removal or grafts; lengthening is apparent and straightening is real.
- **Adolescent severe dorsal chordee: mobilise the neurovascular bundles, incise and graft** — better long-term than incision and cavernostomy. Incision with rotation is reserved for older boys with marked chordee; with rotation alone the bundle stays in its bed.
- **Long-term:** some length gain and a relatively straight penis of adequate calibre for voiding and ejaculation, with minimal morbidity (disassembly also gives a reasonably straight penis).

Bladder Neck Repair

TABLE 43.2 URINARY CONTINENCE AFTER FUNCTIONAL BLADDER CLOSURE

Series population	Dry by day for more than 3 hours
MSRE, males	70%
MSRE, females	74%
YDL after newborn CPRE	56% (lowest)
Repeat closure after failed primary closure	77% (depends heavily on continent diversion)
Large long-term cohort, all continence procedures	76%

Data from Baird et al. (2007); Purves et al. (2008); Schaeffer et al. (2011); Kasprinski et al. (2020); Maruf et al. (2020).

- **What it shows:** after staged repair about **three-quarters are dry by day** in both sexes; CPRE followed by YDL does worst.
- **Trap:** the high rates after failed closure and in the large cohort are **largely achieved through bladder neck closure and a continent catheterisable stoma**, not urethral voiding. In the large cohort, stoma-based closures had the highest continence, **under a quarter voided per urethra**, and nearly a third needed two or more procedures.
- **Other series:** European series with late closure and epispadias repair *after* bladder neck repair did worse (under half dry). Long-term Young-Dees continence was higher in males than females. North American YDL series report **60–82%**.
- **The most important determinant of success is bladder capacity at bladder neck reconstruction.**

TABLE 43.3 URINARY CONTINENCE AFTER 67 INITIAL BLADDER NECK RECONSTRUCTIONS

Result (definition)	Share
Continent — dry day and night, dry interval of 3 hours or more	70%
Social continence — dry 3 hours by day, occasional wet nights	10%
Wet — dry under 3 hours by day, wet at night	19%

- **Context (MSRE males):** closure at a mean of 4 months; bladder neck repair at about 4.8 years; **mean pre-repair capacity about 100 mL**. Continent patients void without augmentation or catheterisation. About 1 in 5 failed completely (continent diversion or awaiting surgery).
- **Postoperative issues:** prolonged outlet obstruction needing a catheter or suprapubic drainage was fairly common. Renal units were preserved; isolated reflux, pyelonephritis and a reimplant revision occurred.
- **Mean time to dryness:** **daytime about 14 months, night about 23 months**. **Age at bladder neck repair did not predict age at continence.**
- **Continence more likely after closure before 72 hours, or after 72 hours with osteotomy. Delayed closure without osteotomy → only about 10% continent.**

- **Capacity over 100 mL** → far higher continence and dryness in **about half the time** (about 10 vs 21 months).
- **Females mirror males** (slightly higher continence; better if over 100 mL). Unlike males, **some girls became continent after bladder and pelvic closure alone**.
- **Timing rules:** delay until **capacity 100 mL and the child is motivated**. Most are dry by day within 12 months; **not dry by 2 years = failure**. Night continence takes 2–3 years. **Desmopressin (DDAVP) or oxybutynin** helps a quarter achieve night dryness.
- **Urodynamics:** continent patients have higher capacity and compliance. Unstable contractions occur in both continent and incontinent groups. **High end-filling pressure is linked to non-obstructive hydronephrosis** → **lifelong follow-up**.

KEY POINTS: PREPARATION FOR CONTINENCE PROCEDURE IN BLADDER EXSTROPHY

- **Bladder capacity over 100 mL (minimum)**
- **Stable urodynamics**
- **Child motivated and ready for a voiding improvement programme**
- **Programme exposure 6 months before** bladder neck repair
- **Evaluation by a child psychologist with a paediatric urology background**

FIGURE 43.27 TIMELINE FOR EXSTROPHY RECONSTRUCTION

Age	Intervention	Considerations
Birth	Bladder closure	Assess plate condition and size; consider osteotomy
6–12 months	Epispadias repair	Assess bladder capacity
2–4 years	Surveillance (yearly cystogram and cystoscopy)	Monitor upper tracts, reflux, infection
5–7 years	Bladder neck reconstruction	Capacity, urodynamics, readiness for voiding programme

Concept — Continence is **decided early and cashed in late**: *template size and a single successful closure set the capacity ceiling, and bladder neck repair at 100 mL or more merely realises it. Failed closures, delay without osteotomy and small capacity are the same failure seen at different times.*

OTHER MODERN EXSTROPHY REPAIRS: CONTINENCE OUTCOMES

CONTINENCE OUTCOMES BY PRIMARY REPAIR

Repair	Continence alone (no bladder neck surgery)	After further procedures	Key caveat
Schrott-Erlangen (single stage)	—	About three-quarters of single-stage patients dry and voiding; about half of those closed elsewhere continent	Some on intermittent catheterisation or augmentation; strict definition (over 3 h, no enuresis)
CPRE	Low (about 1 in 5 or fewer); early dry intervals do not persist	Voided continence about a third; most on clean intermittent catheterisation (CIC) ; about 65–80% need an outlet procedure	Best results: successful closure with osteotomy, hypospadias repair before 1 year, no early reimplantation

Repair	Continence alone (no bladder neck surgery)	After further procedures	Key caveat
Combined closure + epispadias repair (failed or delayed)	A few	About two-thirds of those undergoing YDL dry day and night	Acceptable; but ischaemic penile loss possible; experts only
Kelly	—	Complete continence with unaided voiding under 1 in 5 ; about half partial	Compares favourably with CPRE; glans loss reported

- **Pattern: every “single-stage” repair still needs a bladder neck procedure in most patients.** Continence in CPRE is likelier in females and in primary (not referred) cases, and older age may help.
- **After a complicated newborn CPRE**, few void continently after bladder neck repair; **after an uncomplicated CPRE closure**, most are dry at least by day.
- **Ischaemic penile loss patterns** after combined repair: total glans and corporal loss, hemiglans loss, corporal atrophy, or combined hemiglans and corporal loss.

Concept — *Compressing reconstruction into fewer operations does not remove the need for an outlet procedure; it adds ischaemic and outlet risks. What predicts continence across all methods is the same: an uncomplicated primary closure with the pelvis secured.*

EXSTROPHY RECONSTRUCTION FAILURES AND COMPLICATIONS

Failed Closure

- **Failure forms** (Fig. 43.28): **complete dehiscence, bladder prolapse, outlet obstruction/neourethral stricture, soft-tissue loss, vesicocutaneous fistula** — any of these, or any complication needing reclosure.
- **After failed reconstruction elsewhere**, few reach adequate capacity and only about a quarter become dry → **refer complex cases to expert centres.**

TABLE 43.4 UROLOGIC, ORTHOPEDIC, AND NEUROLOGIC COMPLICATIONS IN 185 PRIMARY BLADDER CLOSURES

Group	Complications	Headline
Urological — major	Bladder prolapse/dehiscence (all males, all reclosed successfully)	About 3%
Urological — minor	Febrile UTI , bladder calculi, posterior outlet obstruction, intrapubic stitch erosion , wound infection, premature suprapubic tube removal	About 11%
Orthopaedic — major (of those osteotomised)	Non-union, leg-length inequality, persistent joint pain	About 6%
Orthopaedic — minor	Pin-site infection, pelvic osteomyelitis, pressure sore	About 3%
Neurological	Femoral nerve palsy	About 2%

- **What it shows:** major failure of closure is uncommon in expert hands. The **commonest problems are minor urological ones**, and the osteotomy itself carries a small major orthopaedic cost.
- **Causes of dehiscence:** incomplete pelvic diaphragm mobilisation, inadequate immobilisation, wound infection, abdominal distension, tube malfunction → **wait 4–6 months before reclosure.** After CPRE it may come with glans, corporal and plate loss (Fig. 43.29).
- **Reclosure principles:** tension-free, **osteotomy**, immobilisation, and **radical dissection of the urogenital diaphragm fibres** (intact in most once- and multiply-failed cases, regardless of age, sex, osteotomy or closure type).
- **Prolapse counts as failure** → reclosure or revision. **Voided continence after reclosure is only about 1 in 6.**

- **Significant prolapse or dehiscence** → at secondary closure **combine epispadias repair** with bladder, posterior urethral and abdominal closure, plus osteotomy, after testosterone (5 and 2 weeks). **Major soft-tissue loss after CPRE** → plan buccal or full-thickness grafts and tissue expansion first. Combined reclosure gives respectable results (complications mostly fistula and stricture) and may avoid augmentation.
- **Neourethral stricture** is associated with **paraexstrophy skin flaps, pubic suture reaction or erosion, and urethral stents**. **Warning signs:** UTI, enlarging bladder on ultrasound, stones, prolonged dry intervals, **unexplained rectal prolapse**. Most occur **within 60 days** of closure.
- **Diversion for obstruction: more than 6 months** → **the bladder usually ends in augmentation; under 6 months** → **usually bowel-free**. Complete outlet obliteration is seen after CPRE and Kelly repairs (ischaemic or technical).
- **Outlet obstruction threatens the upper tracts** (all have VUR) → urethral dilation or incision → open urethroplasty → upper-tract diversion. **If renal function is compromised, secure unquestionable free drainage. No bladder neck or urethral surgery until the stricture is fixed.**
- **Vesicocutaneous fistula often means a failed closure** — a lower midline tract, **diastasis wider than before closure**, and an anterior bladder just beneath the skin on cystoscopy.
- **Successful secondary closure still gives smaller, slower-growing bladders** than successful primary closure. Multiple closures, prolapse, dehiscence, calculi, recurrent infection and vesicostomy all harm bladder potential. After failed CPRE only about half reach bladder neck capacity and a quarter are continent. After repeat closure only a minority are candidates for bladder neck repair and **under 1 in 5 void continently**.
- **Bulking (collagen, dextranomer) after reclosure** to raise resistance and stimulate growth has been disappointing. **No growth** → **augmentation**; problematic neck or urethra → **continent catheterisable stoma** ± bladder neck plasty or transection, with augmentation.
- **Cost:** a failed primary closure adds substantial hospital and professional fees in the following year.

KEY POINTS: EVALUATION OF A FAILED CLOSURE

- Quality of the bladder template
- Perivesical and periurethral scarring
- Pelvic diastasis
- Prior pelvic osteotomy
- **Minimum 4 months' delay since failure**

Failed Bladder Neck Repair

- **Causes:** (1) inadequate outlet resistance; (2) small capacity (no growth after repair); (3) poor compliance; (4) a combination.
- **Failure = no 3-hour dry interval within 2 years.**
- **Near-miss (dry more than 2 hours) → urethral bulking.** Collagen needed repeated injections; dextranomer works better in epispadias than exstrophy.
 - **Before bladder neck reconstruction:** helps half reach adequate capacity but **few become dry**; capacity predicts later success.
 - **After bladder neck reconstruction:** partially dry (1–3 h) → socially continent; **dry under 1 hour → no benefit**. Bulking works only if open surgery has achieved a reasonable dry interval.
 - Prior dextranomer does not make later bladder neck surgery harder, but **repeat injection after a failed injection does not work**.

- **Autologous myoblast injection** is in a phase I trial.
- **Redo YDL:** only for a **patulous neck, capacity over 100 mL and stable urodynamics** (about half of failures qualify); acceptable continence in this select group.
- **Most failures end in augmentation or continent diversion.** Artificial sphincter needs good capacity (rarely present). Reoperation: **bladder neck transection proximal to the prostate with Mitrofanoff**, or an artificial sphincter.
- **Transection + enterocystoplasty:** high initial continence and most dry on catheterisation after revisions; renal unit loss, non-obstructive hydronephrosis and **bladder stones (about 30%)** occur.
- **Delayed continence at puberty** in some males was attributed to prostatic growth, but the **exstrophy prostate is cloverleaf-shaped and absent anteriorly** (normal weight and volume). Any improvement more likely reflects **pelvic floor maturation**.

KEY POINTS: EVALUATION OF THE FAILED BLADDER NECK RECONSTRUCTION

- Bladder capacity under anaesthesia
- Stable urodynamics
- Re-evaluation by the voiding improvement programme
- Re-evaluation by a child psychologist

Failed Genitourethral Reconstruction

- **Urethrocutaneous fistula** is the common complication (roughly 4–19%). Tortuosity or stricture impeding catheterisation is uncommon.
- **Penile disassembly introduced graver losses:** glans, corpora, penile skin and plate (Fig. 43.29C) — from misadventure, vascular variation or intrinsic difficulty. Salvage: tissue expansion, full-thickness and buccal grafts, and ultimately **radial forearm phalloplasty** (a sensate neophallus, also for cloacal exstrophy with little penile tissue).
- **Older patients (scars, short phallus):** excise scar and close primarily, or use flaps or full-thickness grafts. **Tissue expanders inflated over 6 weeks** in severe cases. Free all scar and suspensory ligament; dorsal dermal graft or ventral plication/rotation for chordee. **Over-aggressive lengthening only denervates and devascularises**, because the anterior corpora are congenitally deficient.

Alternative Techniques of Reconstruction

- **Indications:** plate too small or significant hydronephrosis at presentation, a failed closure leaving a small bladder, or failed continence surgery.

Ureterosigmoidostomy

- **First diversion popularised for exstrophy.** Upper tracts are initially normal, so **normal-calibre ureters can be reimplanted non-refluxing**. Metabolic problems fell with better techniques. Attraction: **no stoma**.
- **Prerequisites:** **normal anal continence** and informed family.
- **Complications:** pyelonephritis, **hyperchloraemic acidosis**, rectal incontinence, ureteric obstruction, **delayed malignancy**.
- **Long-term:** many remain in place and continent decades later with mostly normal renal function, but **colorectal cancer occurs** — **almost all poorly differentiated adenocarcinoma**, typically about three decades after diversion. Rates differ between European series → **lifelong vigilance**.
- **Mainz sigma pouch:** non-refluxing reimplants; abdominal wall reconstructed; bladder kept as a seminal receptacle above the prostate; penis reconstructed later. Continence in a rectal reservoir is excellent.

Normal or slightly dilated tracts with intact anal sphincters → Mainz rectal reservoir; severely impaired renal function → colonic conduit.

- **Any mixing of urine and faeces at any stage → lifelong high cancer risk.** Failed reconstruction with polyposis may add risk. **Yearly renal ultrasound and colonoscopy in adult life.**

Continent Urinary Diversion in the Exstrophy Patient

- **Incontinent diversion is rarely needed. Bladder too small to close or to reclose → non-refluxing colon conduit** (protects kidneys from VUR; undiversion possible later).
- **Criteria for bladder neck reconstruction not met** (small capacity, cannot follow a voiding regimen) → defer. **Failed bladder neck reconstruction → augmentation plus continent diversion**, typically with **bladder neck closure**, often ileal or sigmoid cystoplasty with an **appendiceal channel**. Catheterised continence is achieved in the great majority; **bladder stones are the commonest complication**.
- **Failure modes:** nipple-valve de-intussusception, inadequate appendiceal tunnel, persisting bladder neck incompetence — nearly all repairable. Good outcomes even after many prior operations.
- **Early diversion (5 years or younger)** is driven by upper-tract change and social factors (early wish to be dry, severe hydronephrosis, pyelonephritis). It is safe and gives good continence; all stoma patients are dry on catheterisation.
- **Unlikely to reach capacity for bladder neck repair → move early to continent diversion.**
- **Malignancy after continent diversion** is low unless there are co-existing carcinogens (tobacco, chronic immunosuppression) or the inherent exstrophy risk.

KEY POINTS: CONTINENT URINARY DIVERSION IN THE EXSTROPHY PATIENT — INDICATIONS

- Failed closures that do not grow after reclosure
- Newborn closures whose bladders do not grow
- No dryness or inadequate capacity after bladder neck reconstruction
- Upper-tract decompensation after bladder neck reconstruction or primary closure

Concept — Failure compounds: each failed closure lowers the capacity ceiling, which makes bladder neck failure likelier, which ends in augmentation and a stoma. Management therefore pivots on recognising early which bladders will not grow and diverting them before the upper tracts or the child pay the price.

TRANSITIONAL EXSTROPHY PATIENT: ADOLESCENT AND ADULT CONCERNS

- **Lifelong functional and psychological needs:** sexuality and continence are common to the whole complex; cloacal exstrophy adds neurological and orthopaedic disability.
- **Renal preservation** is good after functional reconstruction but can suffer after diversion. Temporary diversion for decompression is usually followed by undiversion to continent pouches; **long-term diversion is now infrequent**.

CONTINENCE

- **Early era:** cystectomy with ileal conduit or ureterosigmoidostomy. **Ureterosigmoidostomy fell out of favour because of lethal colonic cancer**, although renal function was mostly preserved.
- **Now:** continent catheterisable pouches in some centres; mostly **augmentation cystoplasty with bladder neck reconstruction or closure**. With 10 or more years' follow-up, most show no GFR decline, and declines had remediable causes → **augmentation does not compromise long-term renal function**.

CANCER RISK

- **Exstrophy patients with augmentation develop multifocal adenocarcinoma** disproportionately → an **inherent exstrophy risk**. DNA methylation studies do not show a strong epigenetic mechanism.
- **Smoking and post-transplant immunosuppression increase risk**. **Endoscopic surveillance from 5 years after reconstruction** is suggested, though early lesions are hard to see in an augmented bladder.
- **Faecal incontinence** persisting into adulthood has been reported (not confirmed by others) → keep evaluating.

MALE CONCERNS

Sexuality

- **Main concerns: length, appearance and chordee**. Functional length is gained by degloving, excising all residual scar and releasing suspensory ligament remnants.
- **Skin cover**: tissue expansion (pre-planned, 4–6 weeks) or full-thickness grafting (Fig. 43.30).
- **Residual dorsal chordee**: dorsal grafts (dermis, tunica vaginalis, human acellular dermis — preferred as available, durable and easy). Open the penis dorsally for half its circumference. **Measure with artificial erection before grafting**, because saline leaks through suture lines afterwards and misleads.
- **Cloacal exstrophy**: tiny corpora → **flap phalloplasty**; retain all native corporal and glans tissue as an anchor.
- **After assisted lengthening or neophalloplasty**: penile perception and sexual activity rise; most are satisfied; erectile scores and satisfaction are similar across techniques.
- **Libido and sexual function are normal**; erections are preserved after epispadias repair; orgasm is satisfactory in most; partners are satisfied; absent ejaculation mainly after cystectomy.
- **Trap**: questionnaire studies show **more erectile dysfunction** (worse after multiple continence operations) and lower orgasmic scores, yet **desire and satisfaction match peers**. Most men are sexually active.
- **Adult care**: transition to adult urology. **PSA runs below age norms** (changes may be muted) → **routine digital rectal examination and PSA screening** as for other men, plus prostatic hyperplasia assessment.

Male Fertility

- **Paternity is historically rare**. Recent surveys show about a quarter of men achieve paternity, half of them with **assisted reproductive technology (ART)**. Antegrade urethral milking has achieved pregnancy.
- **Functional closure vs diversion**: normal sperm counts are commoner after **diversion** → **iatrogenic verumontanum injury** during closure or bladder neck reconstruction, plus **retrograde ejaculation**.
- Low ejaculate volume, azoospermia and oligoasthenospermia are frequent; raised follicle-stimulating hormone in some. **Genital reconstruction with urethral closure carries high infertility risk**. **Fewer bladder neck operations** → **better sperm counts**.
- **Gamete intrafallopian transfer (GIFT) or intracytoplasmic sperm injection (ICSI)** → successful pregnancies with **no exstrophy in offspring**.

FEMALE CONCERNS

Sexuality

- **Three concerns: external genital appearance, adequacy of the vaginal opening, uterine prolapse**.

- **Recurrent diastasis** widens the scar, separates the pubic hair and splits the clitoral halves → **adolescent touch-up** (Fig. 43.31): lateral **mons flaps** to the midline; for a scarred flat mons, **lateral expanders over 6 weeks** with pseudocapsules rolled for elevation; clitoral halves then reunited. **Topical oestrogen three times daily for 2 weeks** beforehand.
- **Vaginal orifice vertical and stenotic** (tampons, intercourse difficult) → **inverted-U perineal flap** into the incised posterior wall; **daily dilator** until sexually active. **Vaginal cutback increases uterine prolapse**.
- **Perineal flap vaginoplasty, monsplasty and vulvoplasty sequence** (Fig. 43.31): excise the suprapubic scar with V-shaped rhomboid flap incisions → raise rhomboid flaps; inverted-U perineal incision → incise the stenotic introitus and distal posterior wall → advance and suture the perineal flap → trim and suture the labia minora medially, then to the labia majora → rotate rhomboid flaps to the midline.
- **Uterine prolapse: earlier and commoner** (even in nulliparous women). Mechanism: anterior vaginal os, **posteriorly displaced puborectalis with a deficient anterior part**, wide levator hiatus. **Osteotomy does not prevent it; diastasis width predicts it**. Uterine suspension gives only modest protection against recurrence.
- **Suspension technique: sacral suspension** with acellular dermis or porcine collagen sling sewn from cervix to fundus. **Anterior fixation alone is insufficient once prolapse occurs** → fix both sides of the uterus to the presacral ligaments. Consider **prophylactic suspension** at adolescent urinary reconstruction.
- **Sexual function**: normal desire, mostly normal orgasm, occasional dyspareunia; some restrict activity because of genital appearance → **monsplasty with hair-bearing skin and fat**, at primary closure or with adolescent rhomboid flaps.

Female Fertility

- **Fertility is largely preserved**. The **main complication of pregnancy is cervical and uterine prolapse** (frequent, including during pregnancy).
- **Delivery**: vaginal delivery was used after permanent diversion; **caesarean section after functional closure** spares the pelvic floor and sphincter. Pregnancy succeeds after continent diversion too.
- **Outcomes**: most who try conceive; live births with notable miscarriage; postpartum prolapse, temporary incontinence, fistula, ureteric injury and haemorrhage occur → **high risk for mother and fetus → tertiary centre; planned caesarean section is safest**, with a **urologist present** (most patients now follow this).

QUALITY OF LIFE

- **Health-related quality of life (QoL) is reduced** after reconstruction, mainly in physical-activity and general-health domains; **adolescents score better than children and adults**.
- **Adolescent QoL is comparable with norms**, though parents report impaired adolescent health, family activity and their own distress. Incontinent patients trend lower.

LONG-TERM ADJUSTMENT ISSUES

- **Risk factors for difficulty**: multiple operations, incontinence, sexual dysfunction, altered parental interaction, hospitalisation.
- **No clinical psychopathology per se**; the pattern is **acting-out behaviour rather than depression or anxiety**, poorer school function in younger children, and more acting-out if **continence is achieved after age 5**. No differences by sex, bladder vs cloacal type, continence strategy or gender reassignment → **earlier continence is psychologically beneficial**.
- **Adults**: better QoL with continence and good body image. Mostly normative function, with internalising symptoms in older patients and **more impairment in males**. Other work finds **anxiety disorders in all** of a

structured cohort (peer-discovery anxiety wanes after continence) and **suicidal ideation rising with age in males** → **screen for psychopathology as patients age**.

- **Females** have more friendships and partnerships than males; education and careers are good in both. Sexual anxiety is common; **almost all adolescents want psychological help**.
- **Recommendation: early counselling as standard** and psychological support continued into adulthood.

KEY POINTS: BLADDER EXSTROPHY — PRIORITIES IN MANAGEMENT

- Size and quality of bladder template
- Extent of pubic diastasis and pelvic malleability
- Need for osteotomy
- Length and width of urethral plate
- Penile size
- Associated anomalies

Concept — *The adult exstrophy patient carries the late costs of childhood choices: diversion trades continence for cancer, bladder neck surgery trades dryness for fertility, and pelvic anatomy guarantees prolapse. Adult care is therefore surveillance of these specific trade-offs, not general follow-up.*

EPISPADIAS

- **Ranges from a mild glanular defect in a covered penis to penopubic epispadias with complete incontinence** in either sex. It is **most often seen as part of bladder or cloacal exstrophy**.

MALE VS FEMALE ISOLATED EPISPADIAS

	Male	Female
Incidence	1 in 117,000 males (German live prevalence of isolated epispadias about 1 in 17,000)	1 in 484,000 females
Classification	By meatal site: glanular, penile, penopubic ; about 70% complete (penopubic) with incontinence	Davis: patulous orifice (vesicular) → dorsal split of most of the urethra (subsymphyseal) → entire urethra and sphincter (retrosymphyseal, incontinent)
Continence link	Proximal meatus → incontinence (penopubic nearly always, penile often, glanular never)	Most severe degree only
Genital defect	Dorsal chordee; short pendular penis identical to exstrophy	Bifid clitoris , depressed glabrous mons, poorly developed labia minora; vagina and internal genitalia normal
Symphysis	Widened (less than in CBE)	Usually closed or a narrow fibrous band
VUR	30–40%	30–75%
Presentation	Obvious at birth	Minimal external change → may present only with persistent incontinence

Male Epispadias

- **Defect in the dorsal urethral wall:** the urethra is replaced by a broad dorsal mucosal strip running towards the bladder. The meatus itself is not deformed; **incontinence depends on how proximal the meatus is** (Fig. 43.32). In penopubic (subsymphyseal) epispadias the outlet may admit a finger → gross incontinence.
- **Outward-rotated innominate bones** with divergent penopubic attachments → short, pendular penis with chordee, **virtually identical to bladder exstrophy**.

- **Goals:** urinary control, and a straight, cosmetically acceptable penis of adequate length that functions for intercourse.

Associated Anomalies

- **Usually confined** to genital deformity, pubic diastasis and a deficient continence mechanism. Renal anomalies are rare (agenesis, ectopia).
- **Ureterovesical junction (UVJ) inherently deficient**, but **VUR is less frequent than in CBE (about 82% vs 100% in one comparison)**, as are inguinal hernias. The pouch of Douglas is less deep, so the **ureter enters more obliquely**.
- **Key decision:** treat as **isolated epispadias or as exstrophy**, judged by an experienced surgeon from **pubic diastasis (over 3 cm)**, quality of the intrasymphyseal bar, sphincteric tissue on cystoscopy, and capacity. **Treated as exstrophy → osteotomy plus standard exstrophy closure.**

Surgical Management

- **Aims:** continence with upper-tract preservation, and acceptable genitalia. **Incontinence management mirrors closed exstrophy.** First cure of male complete epispadias incontinence: Young, 1922.
- **Good capacity** → epispadias repair and bladder neck reconstruction can be done in **one stage**.
- **Small bladder** → **urethroplasty and penile elongation first**, since a small incontinent refluxing bladder is a poor setting for bladder neck repair. After repair capacity rises substantially within about 18 months, and continence after the later bladder neck repair is high.
- **Capacity is the dominant predictor:** adequate vs inadequate pre-repair capacity gives markedly different continence. **Most become continent within 2 years**, well before puberty (**continence is unrelated to puberty or prostatic growth**).
- **No osteotomy usually** — a firm intrasymphyseal band bridges the gap. At **about 100 mL (typically 4–5 years): YDL plus Marshall-Marchetti-Krantz suspension plus reimplantation.**
- **Genital steps** (as in exstrophy): release chordee and divide suspensory ligaments → free the corpora from the inferior rami → lengthen the urethral groove → lengthen the corpora if needed (incision and anastomosis, grafting, or medial rotation of the ventral corpora downward).
- **Urethroplasty options:** transverse island flap; urethra placed between and beneath the corpora (Cantwell, Ransley); **Mitchell complete disassembly** (reliable chordee correction and preserved erections); disassembly with **perineal muscle reapproximation** may give continence without bladder neck repair. In isolated epispadias, disassembly causes **no glans or corporal ischaemia** (unlike exstrophy), but **often creates hypospadias** that is hard to repair. Significant corporal loss has followed disassembly or Kelly mobilisation.
- **Modified Cantwell-Ransley:** excellent results; **cavernocavernostomy reserved for severe chordee, especially in older boys.** Fistula is uncommon (low single figures to about a fifth early), stricture rare; the neourethra is always catheterisable. Primary repairs have fewer fistulae than secondary repairs. Postpubertal disassembly also gives good cosmesis with a terminal meatus.
- **Long-term:** normal upper tracts; cosmetically pleasing genitalia; normal erections; **most have satisfactory intercourse**; paternity occurs. About two-thirds have a straight, downward-angled penis.
- **Why epispadias outperforms exstrophy:** capacity rises more after urethroplasty, bladders are **more supple and easier to mobilise**, and continence comes sooner after bladder neck repair — because the **bladder was never exposed in utero and never scarred by closure.**
- **Injectables** (plastic microspheres after Cantwell-Ransley; collagen for stress incontinence or as an adjunct after bladder neck reconstruction) help some; complete dryness is uncommon.

TABLE 43.5 URINARY CONTINENCE AFTER BLADDER NECK RECONSTRUCTION (BNR) IN PATIENTS WITH COMPLETE EPISPADIAS

Sex	Continence after BNR (range across six series)
Males	47–82% (highest in the single-centre modified YDL series)
Females	70–87%

Data from Ben-Chaim et al. (1995); Gearhart et al. (1993); Kramer and Kelalis (1982); Arap et al. (1988); Burkholder and Williams (1965); Braga et al. (2008).

- **What it shows:** after BNR, continence in complete epispadias is **at least as good as in exstrophy**, with **girls doing slightly better**; most series used modified YDL. The single-centre female series reports 87.5% in the text and a slightly different figure in the table because the denominators differ.

Female Epispadias

- **Davis classification** (Fig. 43.34) is in the comparison table above. **Mons:** glabrous skin over variable fat, or skin applied directly to the symphysis. **Labia minora** end anteriorly at each hemiclitoris, sometimes with a preputial rudiment. The urethra is seen on minimal labial separation (Fig. 43.33).

Associated Anomalies

- **Deficient UVJ:** laterally placed ureters with a straight course → reflux. **No outlet resistance → small, thin-walled bladder;** reconstruction adds resistance and lets it grow to capacity for bladder neck repair.

Surgical Objectives

- **Continence, upper-tract preservation, and functional, acceptable genitalia**, as in males, guided by an experienced surgeon's judgement.

Operative Techniques

Sequence (Fig. 43.35): lithotomy; urethral roof cleft between 9 and 3 o'clock, blending with hairless mons skin → vertical incision from the base of the mons carried through the **full thickness of the urethral wall at 9 and 3 o'clock** → traction sutures; **excise the urethral roof to near the bladder neck** (often under the symphysis), tapering with a dorsal wedge → **inverting closure over a 10-Fr Foley** from the bladder neck distally → denude the medial hemiclitoris and labia minora → mons fat covers the suture line and fills the space in front of the symphysis → unite the clitoral halves and labia minora (6-0 polyglycolic acid); corpora may be partly detached from the pubic ramus (adds resistance) → create the clitoral hood → lateral subcutaneous mobilisation fills the mons depression (4-0 polyglactin) → skin (6-0 polypropylene) → **catheter 5–7 days** (none if bladder neck reconstruction is simultaneous, when the patient is turned supine for the abdominal stage).

- As in males, the surgeon decides whether to follow an **isolated epispadias or an exstrophy pathway**.
- **Historical procedures failed** (transvaginal plication, muscle transplants, urethral twisting, cauterisation, bladder flaps, Marshall-Marchetti suspension): they raise resistance but **do not correct the anatomy**.
- **A third of incontinent epispadias bladders are under 60 mL.** Augmentation, polytetrafluoroethylene injection and combined neck reconstruction with augmentation have been tried. **Primary urethral closure increases capacity without hydronephrosis.**
- **Timing: urethral and genital reconstruction at 6–12 months; bladder neck reconstruction at 4–5 years** (capacity growth and toilet-training readiness).
- Small series: genitoplasty with percutaneous bladder neck suspension (the neck moved intra-abdominally) had mixed results; urethroplasty with double-breasted perineal musculature gave impressive continence.

Surgical Results

- **Continence about 87.5%** in the authors' girls, comparable with other series of adequate-capacity bladders. All primary patients exceeded 80 mL; after prior urethral and genital reconstruction, **mean capacity at bladder neck repair about 120 mL** → no augmentation or catheterisation needed.
- **Time to continence:** about **18 months** after one-stage genitourethral plus bladder neck repair vs **23 months** when urethroplasty came separately → no speed advantage for staging, but **the capacity gain outweighs it**. Delay may reflect pelvic muscle development.
- **Female epispadias vs CBE:** epispadias girls are reconstructed later, but **bladder growth, procedure numbers, continence and need for diversion are the same**.

KEY POINTS: MALE AND FEMALE EPISPADIAS — PRIORITIES IN MANAGEMENT

- Degree of epispadias and continence
- Associated anomalies
- Bladder capacity
- Aesthetic and functional genitalia

Concept — *Epispadias is exstrophy without the exposed bladder: the genital, pelvic and sphincteric problems are the same, but an unscarred, never-everted bladder grows better, so the same staged logic (urethroplasty first to build capacity, bladder neck at about 100 mL) works at least as well.*

CLOACAL EXSTROPHY

- **Primarily an anterior abdominal wall defect; incidence 1 in 200,000–400,000.** Sex ratio previously 2:1 male, but **probably 1:1**.
- **Mostly sporadic** (occasional unbalanced translocations; sibling recurrence suggests multifactorial causes). Associations: **ART, preterm labour, multifetal pregnancy, maternal smoking**. Mothers were *more* compliant with preconception folate.
- **OEIS complex = Omphalocele, Exstrophy, Imperforate anus, Spinal defects.**
- **Survival is now into adulthood** → focus has shifted to QoL.

Anatomic Considerations

- **Classic constellation:** bladder exstrophy as **two hemibladders**, complete phallic separation, wide diastasis, **exstrophied terminal ileum between the hemibladders**, rudimentary hindgut, imperforate anus, omphalocele; often spinal and lower-limb defects. Urogenital anomalies resemble CBE but are more severe.

CLASSIC BLADDER EXSTROPHY VS CLOACAL EXSTROPHY

Feature	CBE	Cloacal exstrophy
Incidence	About 1 in 46,000	1 in 200,000–400,000
Spinal defects	Uncommon, rarely significant	85–100% (lumbar myelodysplasia commonest)
Omphalocele	Rare, small	Near-universal
Diastasis	Mean 4.8 cm	About twice CBE, asymmetric
Upper-tract anomalies	Uncommon (duplex commonest)	41–60%; about 18-fold CBE; solitary kidney commonest (about 30-fold CBE)
Osteotomy	Selective	Always at bladder closure

Feature	CBE	Cloacal exstrophy
Continence route	Mostly urethral voiding after staged repair	Mostly augmentation plus CIC

Neurospinal Abnormalities

- **Spinal cord or vertebral anomalies in 85–100%: lumbar myelodysplasia about 80%**, thoracic about 10%, the rest sacral. Types: **lipomeningocele (commonest)**, myelomeningocele, spina bifida, isolated tethering.
- **Image every patient:** MRI suggested for all; **spinal ultrasound is comparable in infants**. Only about 1 in 5 dysraphisms shows a sacral skin sign.
- **Mechanism:** disruption of dorsal mesenchyme (or mechanical pulling apart of cord and vertebrae), **not neural tube non-closure**.
- **Function ranges from near-normal to wheelchair-bound. A significant neurological deficit sharply reduces the chance of voided continence.**
- **Autonomic bladder innervation arises more medially** → at risk during dissection → neuropathic bladder after reconstruction. Corporal nerves arise from the sacral plexus, run in the midline, perforate the pelvic floor and pass medial to the hemibladders (Fig. 43.36: also caval duplication). Neural elements are histologically abnormal compared with CBE.

Skeletal System Abnormalities

- **Universal.** The **most extreme diastasis** in the complex: posterior segment angled further back, greater anterior external rotation, **asymmetry**, sacroiliac malformation. **Bone segment lengths are similar to CBE** and bone histology is normal (growth potential normal) → **most need osteotomy**.
- **Other:** vertebral anomalies without myelodysplasia; limb absence or shortening; **clubfoot (commonest limb defect)**; tibial and fibular deformity; congenital hip dislocation; excess hip abduction.

Intestinal Tract Abnormalities

- **Omphalocele in 88–100%**, containing small bowel and/or liver → **close in the neonatal period** to prevent rupture.
- **Associated gastrointestinal anomalies common** (malrotation, duplication, anatomically short bowel equally). **Hindgut remnant** in most.
- **Short gut syndrome in about a quarter, even with normal small-bowel length** (absorptive dysfunction) → **preserve all large bowel**. An unused hindgut is kept for urogenital reconstruction. Modern techniques have reduced short gut.

Genitourinary Abnormalities

- **Müllerian anomalies in most females: uterine (commonest) and cervical duplication**, often bicornuate; vaginal duplication, agenesis, or atresia/hypoplasia; rarely lateral vaginal displacement; complete uterine and tubal duplication. Imaging may miss structures found at surgery. Treatment includes vaginoplasty, septal incision, colporrhaphy and hysterectomy. **Preserve all müllerian duplications** for lower-tract reconstruction.
- **Upper-tract anomalies 41–60%; males more often; abnormal müllerian anatomy → about tenfold renal anomaly risk** → mesonephric duct dysfunction as a common culprit. **Solitary kidney commonest**, then ectopic and malrotated; less often multicystic dysplastic, UVJ obstruction, duplex and fusion. Pelvic kidney, agenesis, hydronephrosis and hydroureter in up to a third. **Ectopic ureters** to vas, uterus, vagina or tube; duplication, congenital stricture, megaureter.
- **Male genitalia:** completely separated, often asymmetric phallic halves and split scrotum; testes often undescended; inguinal hernias common. **Females:** widely divergent clitoral halves.

- **Lower tract:** two hemibladders flanking the bowel segment, **each draining its ipsilateral ureter** and related to the ipsilateral hemiphallus — but **anatomy is unique in every patient**.

Additional System Anomalies

- **Life-threatening cardiopulmonary anomalies are rare** (cyanotic heart disease, aortic duplication, bilobed or atretic lung); caval duplication.
- **Manzoni coding grid** (Fig. 43.37) describes each patient's components systematically (everted bowel, hemibladders, hindgut, hemiphallus, omphalocele) and separates standard from variant forms for planning.

Cloacal Exstrophy Variants

- **Standard form:** everted bowel between hemibladders, omphalocele, blind tailgut with imperforate anus, wide diastasis, bifid phallus, split scrotum or labia; **bifid vagina with two uteri**. Treatment has moved from palliating an almost universally fatal disorder to complex reconstruction with a near-normal lifespan.
- **Variants** (mostly skin-covered; some with bowel and hemibladder duplication) need combined **genitograms, retrograde ureterograms and bowel continence studies** to define anatomy. They have **fewer spinal anomalies and better faecal continence** (pelvic floor innervation permits a Peña pull-through) and fewer comorbidities → treatment can markedly improve QoL.
- **Intravesical phallus** (Fig. 43.38): a rare variant in genotypic males — paired, single or corporal-like tissue anywhere on the bladder plate (usually caudal), with erectile-type histology. **Apparent aphallia → inspect the plate and biopsy any phallic structure before resecting**.

Prenatal Diagnosis

- Early ultrasound criteria: **large infraumbilical wall defect, lumbosacral myelomeningocele, non-visualised bladder**. Only **about 15% are diagnosed prenatally**.
- **Mean gestational age at diagnosis about 22 weeks** (range 15–32). Criteria are graded by frequency, not severity (major = present in over half of cases).

PRENATAL ULTRASOUND CRITERIA FOR CLOACAL EXSTROPHY (AUSTIN)

Major (in more than half of cases)	Minor
Non-visualised bladder (commonest)	Lower-limb defects
Large infraumbilical midline defect or cystic anterior wall structure	Renal anomalies
Omphalocele	Ascites; widened pubic arches
Myelomeningocele	Narrow thorax; hydrocephalus; single umbilical artery

- **“Elephant trunk” sign:** prolapsed terminal ileum as a wavy, cord-like soft-tissue protrusion below the umbilicus — proposed as an added criterion.
- **Fetal MRI** complements ultrasound, is **superior when a normal bladder is not seen**, and defines omphalocele, genital ambiguity and spinal defect better.
- **Purpose:** counselling with psychological support, and **planned maternal–fetal transfer** to a specialist centre.

Concept — *Cloacal exstrophy is CBE plus the neural tube, the gut and the kidneys: every system in the classic form is present and worse, and the added neurological deficit, not the bladder, sets the ceiling on function.*

SURGICAL RECONSTRUCTION OF CLOACAL EXSTROPHY

- **First successful reconstruction:** 1960. Survival was about 50% in the 1970s and **almost 100% by the 1980s** with intensive neonatal care, but is **compromised without subspecialty access**.

Evaluation and Management at Birth

- **Medical stabilisation first**, then a full anatomical inventory for short- and long-term plans, **including gender assignment**. Keep bowel and bladder moist under plastic.
- **Neurospinal defect → immediate neurosurgical evaluation**. Consult social work and paediatric orthopaedics.
- **Gender assignment team:** paediatric urologist, paediatric surgeon, paediatrician, paediatric endocrinologist, child psychologist or psychiatrist, with parental counselling and involvement.
- **Medical concerns or hemibladders too small → intestinal diversion first, delayed closure.**

KEY POINTS: EVALUATION OF THE NEWBORN WITH CLOACAL EXSTROPHY

- Neurospinal anomalies
- Size of the omphalocele
- Size and quality of the bladder templates
- Width of pubic diastasis
- Other associated anomalies

Gender Assignment

- **Historical policy: universal female assignment of 46,XY infants** (small, widely separated corpora) with bilateral orchidectomy, clitoral reconstruction and vaginoplasty.
- **Evidence against it:** reassigned 46,XY children show a **marked male shift in psychosexual development despite no pubertal hormones**; one converted patient virilised from an ectopic testis.
- **Evidence that is mixed:** no difference in social, behavioural or psychological problems by assignment; masculine childhood behaviour with feminine identity in some.
- **Current practice: assign gender congruent with karyotype where possible** (Fig. 43.39; most paediatric urologists favour this). It is supported by **normal neonatal testicular histology** (even when cryptorchid) and by modern **phalloplasty**.

BOX 43.1 MODERN FUNCTIONAL RECONSTRUCTION OF CLOACAL EXSTROPHY

Phase	Steps
Immediate neonatal assessment	Evaluate associated anomalies; decide whether to proceed with reparative surgery
One-stage repair (few associated anomalies)	Excise omphalocele → separate cecal plate from bladder halves → join and close bladder halves with urethroplasty → bilateral anterior innominate + vertical iliac osteotomy → terminal ileostomy/colostomy → genital revision if needed
Two-stage repair — first stage (newborn)	Excise omphalocele → separate cecal plate → join bladder halves (→ bladder exstrophy) → gonadectomy in a male with an unreconstructable phallus → terminal ileostomy/colostomy
Two-stage repair — second stage	Close joined bladder with urethroplasty → combined osteotomy → genital revision
Anti-incontinence/reflux (5–7 years)	Capacity over 100 mL (a very small select group) → YDL + bilateral Cohen reimplants; otherwise augmentation with bowel and/or stomach; continent diversion with abdominal or perineal stoma

Phase	Steps
Vaginal reconstruction	Construct or augment with colon, ileum or full-thickness skin graft

- **What it tests:** the **two-stage route converts cloacal to classic exstrophy first**; a formal YDL is available only to the rare patient reaching 100 mL, and most need augmentation or continent diversion.

Immediate Surgical Reconstruction

- **Individualised plans.** Myelocystocele → **neurosurgical closure once stable**, then lifelong watch for **cord tethering** (symptomatic in up to a third). Spina bifida is common; most walk-age children ambulate fully, some with aids, a few use wheelchairs.
- **Neonatal omphalocele closure** combined with **intestinal diversion**.
- **Hindgut:** formerly ileostomy with hindgut resection; now **always use the hindgut** for fluid absorption. **Tubularised cecal plate with end colostomy reduces short gut**; the remnant enlarges and absorbs more water. It may be anastomosed iso- or antiperistaltically for formed stool.
- **Anal stenosis (not atresia) → potential for continence → pull-through.** An unused hindgut stays as a **mucous fistula** for later augmentation or vaginoplasty.
- **Gastrointestinal reconstruction with bladder closure** → pubic approximation, usually with osteotomy. Some defer bowel reconstruction 1–2 years: near-normal colon, or short colon that still forms solid stool → pull-through with bowel management. **Cannot form solid stool → permanent stoma.** Bowel-lengthening procedures improve absorption.
- **Closure not feasible at omphalocele repair → approximate the hemibladders in the midline without further dissection**, converting to bladder exstrophy; abdominal distension then enlarges the plate for later closure.

Role of Osteotomy

- **Indicated in all cloacal exstrophy at bladder closure** (wide diastasis); medically stable infants may be closed early. It **reduces dehiscence and ventral hernia**: significant complications were almost universal without osteotomy and uncommon with it, in otherwise similar groups. **Osteotomy does not improve eventual continence.**
- **Success by type: combined (about 90%) > posterior (about 76%) > anterior (about 61%).** Osteotomy raises the odds of successful closure **nearly sixfold**. Combined osteotomy avoids repositioning and avoids back-closure complications.
- **Extreme diastasis (over 10 cm) → osteotomy → gradual fixator closure over 1–2 weeks** → then bladder and wall closure. Outcomes and complications are comparable with a single-stage combined approach.
- **Immobilisation:** Buck traction and external fixation perform best; **spica is declining**. An **interpubic titanium bar or plate** maintains reduction (Fig. 43.40). Care is needed with asymmetric pelvis and limb anomalies, which make traction difficult.
- **Risk factors for failed urinary reconstruction** (failed vs successful): **closure in the first week, no osteotomy, spica or mummy wrap**. Successful closures used Buck traction with fixation (or Bryant ± fixation), intersymphyseal plates, and staged fixator cranking over 2–3 weeks. Nearly all successful reclosures had osteotomy.
- **Complications:** more orthopaedic complications (mainly superficial pin-site infection) than in CBE, with no excess of grade 3 or higher. External hip fixation is used more and may contribute.

Bladder Closure in Cloacal Exstrophy

Modern Staged Reconstruction

- Follows **bladder exstrophy staging**. **Independent predictors of successful closure: staged closure, pelvic osteotomy, immobilisation over 4 weeks, and closure by a paediatric urologist.**
- **Dual-stage pathway (DSP)** adoption: closure success rose to 100% with a similar complication rate, even though referred patients now have wider diastases.
- **Technique:** approximate the hemibladders posteriorly → free the lateral edges from the wall → join in the midline → **place bladder and posterior urethra deep in the pelvis**. An **acellular dermal patch** reduces interpubic stitch erosion and penopubic fistula. Repair inguinal hernias at closure. Genital reconstruction follows the assigned gender.
- **Genitalia:** postnatal reconstruction congruent with assigned gender. **Adequate unilateral or bilateral corpora → male assignment favoured → epispadias repair about age 1.** Earlier phallic reconstruction disappointed, but **phalloplasty (various flaps) now lets most genotypic males be raised male**. If female assignment is needed, the phallic halves are joined as a clitoris.
- **Vagina:** early in genetic females; **delayed** in converted males; from a preserved hindgut or expanded perineal skin; long-term dilation may be needed.
- **Fixation and traction 6–8 weeks.**

TABLE 43.6 PELVIC IMMOBILIZATION AFTER THE CLOSURE OF CLOACAL EXSTROPHY (FAILURE RATE, %)

Method	All closures	With osteotomy	Without osteotomy
Spica cast	72	50	86
External fixation + traction	9	9	— (none used)

From Benz KS, Jayman J, Maruf M, et al., J Pediatr Surg, 2018. Differences significant.

- **What it shows:** the same pattern as Table 43.1 but **more extreme**. **Spica fails in most cloacal closures, and half even after osteotomy**; external fixation with traction is used only with osteotomy and fails in under 1 in 10. **Trap:** no immobilisation “rescues” closure without osteotomy — without osteotomy only spica was used, and it almost always failed.

Single-Stage Reconstruction

- Single-stage repair resembles CPRE; it is **deferred if the omphalocele is large or medical issues exist**. **Single-stage failure approaches half vs about 15% for staged closure** → staged closure is recommended. Small single-stage series: dry intervals before toilet training, some needing bulking, occasional augmentation.

Abdominal Wall Reconstruction

- **One of the hardest steps:** diastasis is about twice CBE. Options: non-absorbable mesh, component separation, bioprosthetic material.
- **Sequence:** **pelvic osteotomy → relaxing fascial incisions** (preferred over muscle techniques because the musculature is poor) → if still too wide, **bioprosthetic in preference to synthetic** (less chronic infection, superficial infection treatable without explanting, fewer adhesions and fistulae).
- About a third of staged closures need bioprosthetic material, all without early herniation. Those closed without it are younger. Defect size has not been shown to predict need.

Failed Cloacal Exstrophy

- **Primary closure fails in about 31–43%. Staged closure gives about four times the odds of success of one-stage closure.**

- **Failure mode:** mainly **bladder prolapse**, then dehiscence. **Predictors of successful secondary closure:** **older age at reclosure**, osteotomy, **Buck traction with external fixation**.
- **Costs** are large and rise sharply after failure; **secondary closure complication rates are similar to primary**.

Role of Osteotomy in Failed Cloacal Exstrophy

- **The same principles as primary closure:** osteotomy (nearly sixfold odds of success; associated with successful reclosure) plus **secure immobilisation** — **Buck traction with external fixation is the gold standard**, clearly better than spica, which dominates among failed secondary closures. **Recommended:** **combined bilateral anterior innominate and vertical iliac osteotomy with 6 weeks of Buck traction and external fixation; spica is not recommended**.

Techniques to Create Urinary Continence

- **Continence is possible in most but usually needs augmentation and intermittent catheterisation.** A **median of two procedures** (range one to four) is needed for a dry interval over 3 hours, often from revisions of the continent diversion. **Median age at urinary continence about 11 years.**
- **Augment with hindgut (if available), ileum or stomach.** Gender-converted males → **continence hardest → continent stoma**. Genetic females: YDL can work, but most need catheterisation. **YDL success tracks the neurological status.**
- **Components:** orthotopic urethra from local tissue, vagina, ileum, stomach or ureter; catheterisable ileal stoma if bowel length and fluid balance allow. **Delay the continent reservoir until the child can self-catheterise.**
- **Urethra vs abdominal stoma** depends on outlet adequacy, dexterity and interest, and orthopaedic status (spine, hips, braces, ambulation). More than half achieve continence, most via a continent stoma, the rest voiding or catheterising urethrally.

Concept — *In cloacal exstrophy closure success is an orthopaedic problem and continence is a neurological one: staging, osteotomy and rigid fixation secure the closure, but the spinal lesion and the gender-assignment path decide whether the child voids, catheterises a urethra, or uses a stoma.*

LONG-TERM ISSUES IN CLOACAL EXSTROPHY

- **Survival is almost universal** → the aim is QoL through function, with **individualised management**.
- **The main determinant of long-term disability is the level of the neurological defect** → early aggressive neurological management and **lifelong tethering surveillance**. Minimal deficit → aim for **bowel pull-through and voided continence**.
- **Ricketts continence score** (6-point bowel plus bladder; 6 best, 0 worst): most children score poorly (colostomy with an incontinent bladder); very few reach an enema programme with a continent bladder.
- **Permanent ileostomy vs terminal colostomy:** ileostomy has more early morbidity, but **bowel adapts by about 3 years**, and with aggressive parenteral nutrition growth is similar. Modern bowel reconstruction avoids long-term short gut in most.
- **Phallic reconstruction:** past lengthening failed (tiny corpora, wide separation) → **neophalloplasty** (forearm or other flap). Satisfaction with penile reconstruction is about 60% and does not differ by method.
- **Fertility:** **male fertility compromised** across the complex, although paternity occurs spontaneously or with ART. **Female fertility typically preserved** with little ART → **caesarean section with a urologist present**. **Cervical prolapse is worse than in bladder exstrophy.**

SUMMARY

- **Near-universal survival** with major improvement in cosmetic and functional outcomes. **Gender reassignment remains debated**; long-term data are accruing. **Continence is achievable** with reconstruction and intermittent catheterisation; many patients lead fruitful lives.

KEY POINTS: CLOACAL EXSTROPHY — PRIORITIES IN MANAGEMENT

- Medical stabilisation
- Evaluation of associated anomalies
- Colonic functionalisation
- Separation of bladder from the gastrointestinal tract
- Functional genital reconstruction